

Harness Credit Is Mostly Proximally Mediated: Four-Arm Counterfactual Replay Separates Total from Direct Effects in a Two-Layer LLM Agent

Anonymous authors
Paper under double-blind review

Abstract

An agent built on a large language model (LLM) is the model wrapped in a harness that manages its context, retries, and termination. Both layers decide, but outcome-level reinforcement learning cannot tell their contributions apart, and single-layer credit bills one layer for effects that may belong to the other. We measure the split with deterministic counterfactual replay (noise floor exactly zero) on a coding agent, completing the 2×2 factorial grid at each decision pair — including the cell prior work omits: the harness flip with the model’s original next action held fixed. A context decision never touches the environment, so some model action must carry its effect; the question is which. On the stacks we measured, the very next one does: at 36/37 reward-pivotal slack points the harness decision’s total effect is large while its effect with the next action fixed is exactly zero, and the cell replicates on MBPP+ and HumanEval+ (48/49). Three quarters of those points have no later write, where this is the expected reading; where a later write could have intervened, 21/23 still mediate proximally. Mediation turns distal on a larger model and for later consumption of the destroyed information, and is absent for termination decisions, which have a direct path by construction. Consequences: the three-arm “interaction” $C_{HM} - C_H - C_M$ is a total-minus-direct contrast; C_H is the estimand for harness-only training, where exact but sparse replay credit did not beat outcome-only at matched dose; joint training needs direct effects, which are zero where the mass is. We release the apparatus and a 43-item pre-registration ledger.

1 Introduction

An agent built on a large language model (LLM) is not a model. It is a model wrapped in a harness: a program that manages context, retries failed calls, and terminates episodes. The harness makes as many decisions per episode as the model does, and those decisions — especially context management — destroy or preserve the information the model conditions on. Yet outcome-level reinforcement learning (RL) assigns one scalar return to every decision regardless of layer, and per-step credit methods attribute within a single layer only. When a harness decision matters, how does it matter: directly, or through the model action it changes? The two readings bill different layers.

We answer with deterministic counterfactual replay: re-execute a recorded trajectory from any decision point with exactly one decision changed and the prefix forced, defining per-decision credit $C(d) = R_{\text{orig}} - R_{\text{cf}}$. Because replay is exactly deterministic (measured, not assumed), every nonzero credit is signal. At a paired point — a harness decision h at turn i followed by the model action a it conditions at turn j — we measure all four cells of the 2×2 grid. C_H (harness credit) flips h and lets the model respond live: the total effect of the harness decision. C_M (model credit) forces one alternative action a' under the factual h ; C_{HM} forces both. The cell prior work omits, $C_{Ha} = R - R(h', a)$, flips h with the model’s original next action held fixed: the harness effect not carried by that action. In causal-mediation vocabulary (Pearl, 2001), with h as treatment and a as mediator, C_H is

the total effect, C_{Ha} a direct effect with the mediator at its factual value, and $C_H - C_{Ha}$ the part transmitted through the very next model action (Remark 2). Forcing an action is a do-intervention, not a prediction, so “the model’s action depends on the harness” is the design’s premise, not an objection to it.

The finding: harness credit is mostly proximally mediated, and the stratification says how much of that is measured. For a context decision the question is not whether the model carries the effect — context operations never touch the environment, so some model action must (Corollary 1) — but which: the very next action, or ones further down the trajectory. On our primary triple (Qwen3-4B, a context-managing harness, designed coding tasks with budget slack), the harness decision has a large total effect at 37 reward-pivotal points (median $|C_H| = 0.875$), and at 36/37 of them $C_{Ha} = 0$ exactly: the whole effect passes through the next model action (pre-regs 40–41, §4.4). A re-sampled next action under the flipped context reproduces R_H at 36/44 points, and on two external benchmarks the fourth cell replicates at 48/49 pivotal points (pre-reg 42, failure branch declared in advance). How much of this the design entails is itself measured (pre-reg 43): at 63 of those 84 points the paired action is the episode’s last write, so $C_{Ha} = 0$ is the expected reading; at the 23 points where a later write exists, 21 still mediate proximally. Mediation turns distal on the 8B (2/25 at such points) and is absent for termination decisions, which end the episode and hence have a direct path by construction (10/10, a positive control). The three-arm quantity $I = C_{HM} - C_H - C_M$ of our earlier registrations (and of single-layer replay work) is therefore a total-minus-direct contrast, $I = I_{\text{fact}} - (C_H - C_{Ha})$ (Remark 3), with the factorial interaction I_{fact} zero at 56/57 slack points.

What this implies for training. If the harness is the only trainable layer, C_H — total effect, model at its live policy — is the correct estimand, and outcome-only policy gradient is its Monte Carlo estimate without the replay tax; under dose-matched budgets the exact signal lost to outcome-only in 3/3 seeds (§6) for a measured reason: it updates 39% of decisions with gradients $\sim 5\times$ smaller. Joint training, where the model’s own update absorbs the mediated part, needs direct effects instead — exactly zero where the harness mass is. A census of $C_{Ha} \neq 0$ therefore prices, before training, how much joint-specific signal a stack offers (Corollary 3); on both of ours it priced it near zero.

Contributions.

1. Measurement. Deterministic counterfactual replay with a noise floor measured at exactly zero under explicit serving premises (vLLM’s automatic prefix caching silently breaks it), delivering the complete 2×2 grid per decision pair on the same trajectory.
2. Finding. Harness credit is mostly proximally mediated — carried by the very next model action (36/37 pivotal slack points, 16 unique decisions over 14 tasks; re-sampled action consistent at 36/44; 48/49 on two external benchmarks; 21/23 where a later write could have intervened) — turning distal on a larger model (8B, 2/25) and absent for termination (0/10, positive control). The three-arm interaction is a total-minus-direct contrast; we give the identity relating it to the factorial interaction.
3. Prescription, with its cost measured. C_H for harness-only training, direct effects for joint training, a census gate that prices the latter before training; one dose-matched training act (after three diagnosed failures) plus an episode-matched control show the exact signal losing to outcome-only, with the mechanism (sparsity $\times$ step size) identified.
4. The record. A 43-item pre-registration ledger with failures reported as results, a three-arm misreading corrected by its own fourth arm (pre-reg 40), and an auditable claims table (Tables 3, 9).

This is a measurement paper in the genre of sanity checks and audits (Adebayo et al., 2018; Lucic et al., 2018; Schaeffer et al., 2023; Henderson et al., 2018): the instrument, the identity, and the census gate are the contributions.

2 Related Work

Counterfactual replay for agent credit is an active area, but every existing system we know of attributes within a single layer. CHILL-Harness (Fu et al., 2026) intervenes on harness decisions, training an amortized effect estimator from checkpointed paired replay, but never measures the model layer or the interaction; our single-layer arm reproduces its principle (§6). CAR (Shah, 2026) formalizes per-step do-operations with Shapley attribution under a replay budget — including step-step interaction — but within one decision stream; composing the two would not yield the joint arm C_{HM} or the fourth cell. C3 (Chen et al., 2026a) audits credit fidelity against exact replay ground truth (the validation metric we adopt) but decomposes across agents, not across the model/harness boundary inside one agent. A recent negative result (Zhang, 2026) shows popular step-level credit signals fail to beat chance against replay ground truth, setting the evidential bar our critic section inherits. DFAH (Khatchadourian, 2026) finds trajectory determinism generally unattained; we engineer the zero floor and build causal credit on it. Co-Harness (Chen et al., 2026b), HarnessCompass (Zhang et al., 2026) and HASE (Luo et al., 2026) jointly optimize harness and weights, and Agent Lightning (Luo et al., 2025) provides rollout infrastructure, all without a per-decision causal signal. The correction rule $C_{HM} - C_M$ sits in the lineage of difference rewards and COMA (Wolpert & Tumer, 2001; Foerster et al., 2018), which marginalize one agent’s action against a baseline; we marginalize one layer’s decision against a measured joint counterfactual. Hindsight credit assignment (Harutyunyan et al., 2019) and process reward models (Lightman et al., 2024) amortize per-step credit inside the model layer; the mediation vocabulary is Pearl’s (Pearl, 2001). Table 8 (Appendix H) summarizes the gap: no prior work measures total, direct, model and joint effects at the same decision point against an exact zero-floor replay.

3 Setup: A Two-Layer Agent with Deterministic Replay

3.1 Agent, decision schema and recorder

The agent is deliberately minimal: a sandboxed coding agent that reads a task description, writes files, and runs tests. The model layer (Qwen3-4B served by vLLM 0.8.5.post1 on one RTX 4090, greedy decoding) chooses tool calls: `write_file`, `run_tests`, `read_file`, `finish`. The harness layer is a scripted policy that at every turn makes a `context_policy` decision (`keep_context` vs. `summarize_context`, the latter compressing history to a fixed-length prefix) and a termination decision. Every decision of either layer is recorded with a typed schema (id, layer, state before, available actions, chosen action, observation, token costs, parent/child links). Reward is deterministic: the fraction of hidden pytest tests passed, $R \in [0, 1]$; cost-adjusted reward is $R_{\text{eff}} = R - \lambda \cdot \text{prompt tokens}/10^5$, with λ fixed before each run it governs (Acts 1–3: 25; Act 4 and V2: analytic λ^* of 5 and 0.2; raw token totals are stored).

3.2 Replay and interventions

Replay re-executes a recorded trajectory through the live agent loop with a forced-action queue (Figure 1): recorded decisions are matched (on type and state hash) and forced up to the intervention point; from there the loop runs free. An intervention replaces exactly one decision (or, for C_{HM} , one harness–model pair). Model-layer alternatives a' are sampled once per point by a fixed procedure (temperature 0.8, seeds 2001–2008, first parseable action differing from the recorded one, conditioned on the recorded un-summarized state); every model-layer credit is an estimand relative to this single alternative, with re-sampling sensitivity measured (§4.3). All replays run sequentially against one vLLM instance.

3.3 Tasks

The designed pool has 30 synthetic coding tasks in four strata (training runs split them 20/10; a 22-task curated pool derived from a hardened variant, Appendix D, brings the project total to 52 unique tasks). V2/L tasks place critical constants beyond the first 240

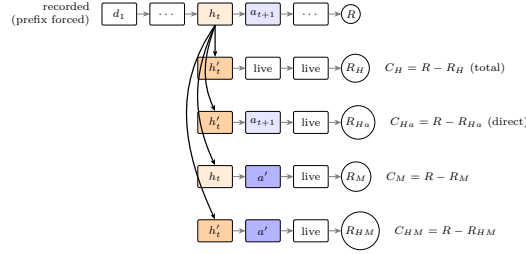


Figure 1: The four-arm replay at one interaction point. The recorded prefix is forced up to h_t ; from the intervened decision on, the loop runs live. H arm: model re-queried under the flipped context (total effect); Ha arm: flip paired with the original action forced (direct effect; fourth cell, pre-reg 40); M and HM arms force the sampled alternative a' . Screening-off is the event $C_{HM} = C_M$.

characters of history, so a summarize decision is informationally destructive at known positions (L are longer variants); S tasks require harness-model cooperation by design; C tasks are controls (all needed information survives any summary, predicting $C_H = 0$). Reward saturation ($R \in \{0, 1\}$) is endogenous to the tests per task (§4.3). The same instrument also runs on two external benchmarks (MBPP+ and HumanEval+, test-hardened program-synthesis suites (Liu et al., 2023)) and on three further models (Qwen3-1.7B/8B, Mistral-7B; §8).

4 Measuring Cross-Layer Credit

4.1 Noise floor

Every causal claim below is only as good as the replay, so we measure the floor rather than assume it: the pre-registered floor per configuration is $\max |\Delta R|$ over null replays (defined on R , not trace equality). Across 26 dedicated null-replay runs — three models (Qwen3-4B, 1.7B, 8B), three environments (designed pool, MBPP+, HumanEval+), the hardened/curated pools, and an LLM-summarizer cell — plus 1,568 nulls attached to the 24 interaction runs, every one of 9,296 null replays reproduced R exactly: the floor is 0.0 in all configurations (Figure 5a; Table 11 reconciles every count against the run artifacts). This is a measurement under explicit premises — fixed serving config, strictly sequential requests, prefix caching off — not a guarantee.

The premises earn their keep: three root-caused incidents show the floor is fragile to any unaudited byte or state — an ASLR (address-space layout randomization) address leaking through a pytest repr (2/432 pre-fix nulls), an accidental concurrent chain ($\Delta R = +0.67$), and a greedy LLM summarizer replacing the rule-based one (floor survives: 317/317 exact). Appendix L reports all three in full. The sharpest premise is prefix caching. vLLM enables automatic prefix caching (APC) by default; with APC on, prefill numerics depend on the key-value cache state left by previous requests, so greedy decoding diverges across identical prompts even when strictly sequential: in our APC-contaminated run 7/270 nulls were inexact and the floor rose to 0.417 (Figure 5b), enough to swamp most true credits. Sequential execution is not sufficient, and any replay-based attribution on default vLLM settings inherits this silent failure. Known serving-nondeterminism work targets batching-induced kernel effects (He & Thinking Machines Lab, 2025); APC is distinct — cross-request state — and we quantify it as a replay hazard, not claim its discovery.

4.2 C_H and C_M

With an exact floor, per-decision credits are directly interpretable: any $C \neq 0$ is a real causal effect of that single decision (248 harness and 66 model points, Figure 6, Appendix H). Direction is everything (keep→summarize at an information-critical point yields C_H up to +0.89, the reverse near zero; we never aggregate across directions); credits are sparse,

Configurations	
gN	slack regime: max 12 turns, summarize threshold N prompt tokens (g450/g600/g900; g600 $\equiv$ mt12)
mtK	pressure regime: max K turns, threshold 600 (mt4/mt6/mt8)
q8 / q17 / none	Qwen3-8B / Qwen3-1.7B / Qwen3-4B (primary)
mbpp / he; cur; ls	MBPP+ / HumanEval+ environments; curated pool; LLM summarizer cell
V1 / V2	single-file harness (context + termination) / mini-SWE harness (five decision types, §7)
CI95	95% confidence interval, task-clustered unless noted
Outcome codes (declared before each run; thresholds per item in Table 9)	
s1 / s2 / s3	prediction holds / partial / fails (screening, replication, training bins); m, k, c, o, X, E, p, d, r, b + digit: cell-specific declared outcomes

Table 1: Glossary of configuration names and pre-registered outcome codes.

concentrated at the designed information-destruction points; and C_M at matched points is frequently nonzero when C_H is not.

4.3 Interaction I and its anatomy

The central measurement flips a harness and a model decision at the same point, alone and jointly. The model action screens off the harness decision when $C_{HM} = C_M$: conditional on the model counterfactual, the harness flip has no marginal effect (term from Reichenbach, 1956; Appendix B). Table 1 fixes the configuration names and outcome codes used throughout; every screening count in the paper, with its endpoint, is in Table 4 (Appendix C).

With budget slack (g450/g600/g900) the model action screens off the harness decision exactly: $C_{HM} = C_M$ in 57/57 points (reward-pivotal subset 53/56, CI95 [0.87, 1.00]; discovery 14/14 and confirmation 39/42; a' re-sampling 99/101). Under pressure (max 4/6/8 turns) screening breaks in 8/65 points ($p = 0.005$ point-level; $p = 0.125$ task-clustered over 3 breaking tasks), at the same points across configurations. The break rate is non-monotonic in pressure (0/21, 4/22, 3/21, 1/22 for mt12/8/6/4; registered monotonicity test failed, $p = 0.38$) and tracks reward saturation: at mt4 the agent fails most tasks outright, so no flip can make things worse (non-saturated points: 0/12 vs. 6/19, $p = 0.037$; post-treatment). We call this a competence window: cross-layer interaction is detectable only where the agent neither saturates success nor failure. Across models and pools (Table 5, Appendix D) pressure-gating is a 4B-on-designed-pool phenomenon: the 8B breaks at a flat rate in both regimes, and the external benchmarks screen everywhere.

Anatomy and consequence (summary). Two verified mechanisms — structural re-injection (a' , sampled from the un-summarized state, carries the destroyed information across the cut; 23/24 annotated) and downstream re-summarization (22/52) — explain 72/75 screened points (Appendix C). At every screened point $C_{HM} - C_M = C_H + I$ gives $C_H = -I$ (Proposition 1): the two candidate harness signals differ by the whole of C_H . Control tasks return $C_H = 0$ in 11/12 points (Appendix B).

4.4 The fourth cell: total vs. direct harness effect (pre-regs 40–43)

The C_H arm is $R(h', f(h'))$ — the model answers the flipped context live — so it is a total effect, and $I = C_{HM} - C_H - C_M$ is not a factorial interaction. An external audit predicted, testably, that at slack screened points the original action a — generated seeing the intact context — re-injects the destroyed information itself, so $R(h', a) = R$. Pre-reg 40 registered three outcomes (s1 ≥ 0.90 / s2 / s3 < 0.50) and ran one greedy replay per point (flip at i , original action forced at j) on all 122 V1 and 48 V2 census points (10 termination duals analytic) after a null gate (16/16 exact). Outcome s1: 56/57 (0.982).

Pre-reg 41 re-read the same rows as a mediation table (Table 2). Among the 57 slack points, 37 are reward-pivotal for the harness ($C_H \neq 0$; the other 20 have $C_H = C_{Ha} = 0$ trivially, so 56/57 and 36/37 count the same rows). At 36/37 of them $C_{Ha} = 0$ exactly while $|C_H|$ is large (median 0.875, IQR [0.84, 0.90], max 1.0): the harness decision changed the reward by nearly a full task, and all of that change is carried by the next model action.

Stack	Population	Pivotal	$C_{Ha} = 0$	...at $H_w \geq 1$	med. H : $C_{Ha} \neq 0$ vs. $= 0$
V1 4B	slack (g450/600/900)	37	36 (0.973)	12/13	8 vs. 1
V1 4B	pressure (mt4/6/8)	46	39 (0.848)	11/18	3 vs. 1
V1 4B	MBPP+ (g600/mt6), pre-reg 42	8	7 (0.875)	5/6	—
V1 4B	HumanEval+ (g600/mt6), pre-reg 42	41	41 (1.000)	4/4	—
V1 8B	designed pool, screened (g600/mt4/mt6)	57	51 (0.895)	2/8	—
V1 8B	designed pool, non-screened	17	0	0/17	shield breaks = direct effect
V2 4B	context_policy	23	13	6/14	6.5 vs. 1
V2 4B	observation_policy	12	4	4/11	12 vs. 15
V2 4B	test_schedule	3	2	—	—
V2 4B	termination	10	0	0/8	direct by construction

Table 2: Mediation table (pre-regs 41–43). Pivotal = $C_H \neq 0$; $C_{Ha} = 0$ = no effect once the model’s original next action is held fixed. H_w = write_file calls after j in the recorded episode; at $H_w = 0$ the paired action is the last write and $C_{Ha} = 0$ is the expected reading (it held at 140/140 such V1 points and was violated at 3 V2 points, where the live suffix under h' added a write). H = tool calls after j ; medians are descriptive — no horizon contrast survives Holm correction together with task-level permutation (Appendix F). Termination ends the episode: a direct path by construction, the typology’s built-in positive control.

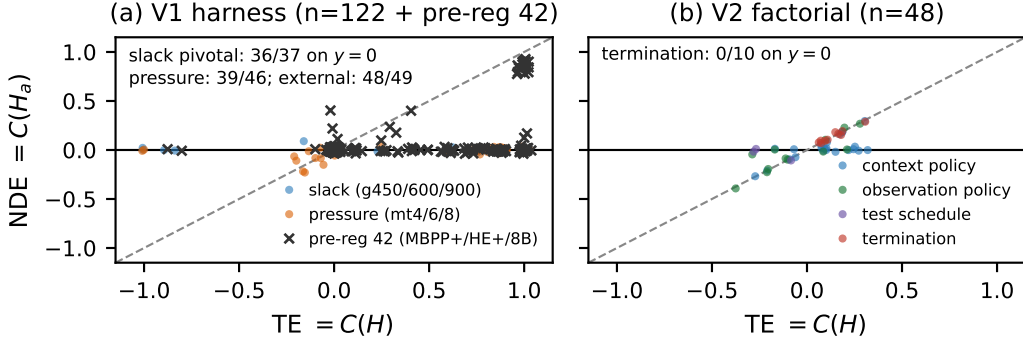


Figure 2: Total vs. direct harness effect at every measured point (pre-reg 40 and 42 rows). (a) V1, 122 designed-pool points by regime plus the 240 pre-reg 42 points (crosses: MBPP+, HumanEval+, 8B): slack pivotal points sit on $C_{Ha} = 0$ (36/37; external 48/49), pressure points start to leave it (39/46), and the 8B’s shield breaks sit near the diagonal. (b) V2, 48 points by type: termination lies on the diagonal $C_{Ha} = C_H$ (no mediation, 0/10); context points split by horizon. Jitter $\sigma = 0.015$.

Because the same decision is measured under g450/g600/g900, the 37 points are 16 unique (task, decision) pairs over 14 tasks (15/16 mediate; the 57 points are 29 pairs over 20 tasks); per configuration 12/12, 12/13, 12/12. The one break (1_log_parser, $C_H = -0.154$, $C_{Ha} = +0.077$) has the information consumed two turns later.

Three secondaries locate where mediation fails (Figure 2). (i) Coupling, not draw. The 44 points where pre-reg 26 had re-sampled the next action under the flipped context (a'_s , temperature > 0) give $R(h', a'_s) = R_H$ at 36/44: a different next action under h' reproduces the live arm’s reward; all 8 discordances have $R(h', a'_s) > R_H$ (a better action found, not a direct path). (ii) Measured vs. expected: the last-write stratum (pre-reg 43). If the paired action is the episode’s last write ($H_w = 0$), $C_{Ha} = 0$ is what Corollary 1 leads one to expect — the default, not a theorem, since the live suffix under h' may add a write. Registered before computation: 63 of the 84 mediating headline points (36 slack + 48 external) sit at $H_w = 0$, where the expectation held at 140/140 V1 points (3 V2 context points violate it: seeing the summarized context, the model wrote more and scored higher). At the 23 headline points with a later write the cell is informative, and 21/23 mediate proximally (slack 12/13, external 9/10; outcome h1). The same stratum separates models: 8B 2/25, V1 pressure 11/18, V2 context 6/14 (Table 2). Because C_{Ha} fixes only the immediate action, $C_{Ha} \neq 0$ at a context decision is distal mediation, not a direct path: C_{Ha} is the distal share,

$C_H - C_{Ha}$ the proximal one. Points with $C_{Ha} \neq 0$ have longer remaining horizons in every population, but only V1 pressure survives Holm ($p_{\text{adj}} = 0.005$) and none survives task-level permutation (Appendix F): descriptive. (iii) Type. Termination decisions — the only type with a direct path by construction — give $C_{Ha} \neq 0$ in 10/10; observation formatting is re-read at every turn (4/12). Consequently I throughout this paper is a total-minus-direct contrast, $I = I_{\text{fact}} - (C_H - C_{Ha})$, and $C_H = -I$ at screened points (Proposition 1) reads: the single-layer estimator bills the harness a total effect whose immediate-direct component is zero (Remark 3). The measurement claim is unchanged; the estimand each arm delivers is now named.

Outside the designed pool (pre-reg 42). Because the designed pool places the destroyed information exactly where the next action needs it, 36/37 could be a design artifact; we registered that failure branch (fraction < 0.60) before touching the data and replayed the fourth cell at every census point of MBPP+ and HumanEval+ (143 points, slack and pressure; null gate 20/20 exact on new hardware). At the 49 reward-pivotal points (29 unique pairs) $C_{Ha} = 0$ in 48/49 (s1; MBPP+ 7/8, HumanEval+ 41/41; median $|C_H| = 0.80$). The one break (mbpp_7, pressure: $C_H = C_{Ha} = 0.4$, $C_{HM} = C_M = 0$) is the V1 pattern: the original action fails to re-inject what the sampled a' does. Non-pivotal points are exact at 91/94; $I_{\text{fact}} = 0$ at 139/143. These cells are saturated ($R = 1$ at 43/49): they test mediation where the census already screens. The secondary 8B cell (97 points, Appendix D) adds the converse: where the 8B breaks screening the fourth cell breaks too ($C_{Ha} \neq 0$ at 17/17 non-screened pivotal points, $I_{\text{fact}} = 0$ exactly at 15), while screened pivotal points mediate at 51/57. Non-screened mass and direct-effect mass are the same points (Corollary 3).

5 The Structure of Harness Credit

Can C_H be predicted from cheap features instead of replayed? Linear and gradient-boosted critics trained on 248 ground-truth points never beat the context-size heuristic mandated as baseline by (Zhang, 2026) (pooled Spearman 0.718 vs. 0.770; no stratum reverses the order; Appendix I): harness credit is mostly a function of context size and position, as the information-destruction mechanism predicts. A C_M critic trained on the designed tasks transfers to 94 MBPP+ points at chance (Spearman 0.03/−0.08; pre-reg 16).

6 Training with Cross-Layer Credit

Does better credit train a better harness? No — and the mediation reading says why (Figure 4; Table 7, Appendix G). Dose-matched arms test whether the billed signal can pay its replay tax (Proposition 2). We train the harness context_policy (a logistic policy over 5 features) with four arms dose-matched in total rollouts (episodes + credit replays): (1) outcome-only policy gradient; (2) single-layer C_H credit (the CHILL-Harness principle); (3) interaction-corrected credit $C_{HM} - C_M$; (4) a zero-credit control (no replays; de facto keep-always). The registered criterion (GATE 3) required arm 3 to beat arms 2 and 4. Acts 1–3 (Appendix G) exonerated the estimator (58/60 sign agreement on fresh replays), repaired the optimization, and exposed the objective as mis-specified: under $\lambda = 25$ universal cheap failure was the true optimum (9/9 runs collapse to summarize-always), so a sane objective must verify the margin against every expressible fixed attractor.

Act 4: sane objective — the arms separate, and the exact signal loses on step size. The last run (outcomes declared before any rollout) repairs the objective: $\lambda^* = 5$ and 12 tasks selected analytically from attractor measurements on all 52 tasks, so that a context-size threshold policy (thr600) strictly dominates both fixed attractors held out (0.455 vs. keep-always 0.392, summarize-always −0.031). For the first time arms 2 and 3 separate, in 3/3 seeds: outcome-only reaches held-out $R_{\text{eff}} = 0.440$ –0.443, single-layer C_H 0.398–0.405, and $C_{HM} - C_M$ stays exactly at keep-always (0.392, the untrained control). Our registered secondary predicted the opposite ordering; its failure is the finding. For a harness-only trainer C_H is the correct estimand (total effect, model at its live policy), and outcome-only policy gradient is its Monte Carlo estimate with no replay tax; $C_{HM} - C_M$ is a direct-effect

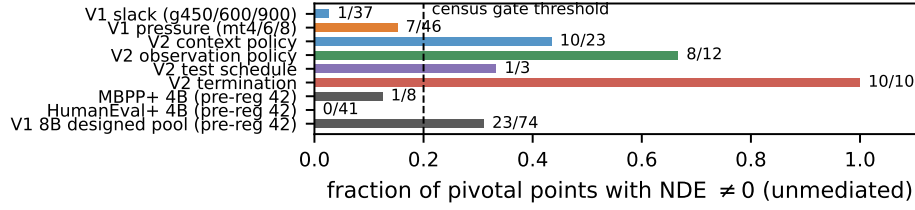


Figure 3: The census in its mediation reading: fraction of reward-pivotal harness points with $C_{Ha} \neq 0$, by population and type (pre-reg 40 and 42 rows). V1 slack and the external benchmarks offer almost no direct-effect mass (1/37, 1/49); the 8B offers it where its shield breaks; V2 in every type, entirely for termination.

quantity, exactly zero at 36/37 slack points, so arm 3 receives no gradient where the harness acts. A registered episode-matched control (pre-reg 31) closed the sample-size confound (credit arms: 139/70 episodes vs. 280): outcome-only stopped at the credit arms’ episode counts still reaches 0.398–0.450, $\geq$ the C_H arm in 3/3 seeds. The training logs supply the mechanism (pre-reg 41e): the C_H arm assigns nonzero credit to 39% of decisions (outcome-only: all), with $|C_H|$ median 0.039 — $20\times$ below the census’s pivotal points — and an episode-summed gradient norm $\approx 5\times$ smaller (0.062 vs. 0.295): at equal learning rate the exact arm moves the policy $\sim 5\times$ less per episode than the dense, noisy outcome signal — an optimization-efficiency gap a re-scaled step could close (untested here), not an estimand error.

7 The Other Branch of the Rule: A Second Agent Class

If mediation were an artifact of a single dominant harness decision, the census would have nothing to price elsewhere; we built a second agent class and ran the census as a decision procedure, its gate registered up front (Corollary 3).

Stack, floor, pivotality. Mini-SWE tasks: a frozen pool of 60 multi-file repos (15 families $\times$ 4 bug variants; 4–6 files, 6–12 pytest tests; buggy state strictly inside (0, 1)). The harness makes five decision types (context policy, observation formatting, test scheduling, retry, termination), each logged and forceable; two configurations (slack/pressure) give 120 baseline episodes. A pilot finding: under a JSON action envelope the model emitted zero write_file actions in 437 calls — the envelope is itself a first-order harness decision; under plain text the a' sampler failed at 87/114 points until a registered escalation to temperature 1.2. The zero floor survives the class change (117/117 nulls exact); 114 reward-pivotal points were found, all five decision types pivotal.

The census verdict. At 48 valid census points (65 lacked an a' even after escalation) the registered outcome is s3, general break: no type screens at ≥ 0.90 — context policy 0.609 ($n = 23$), observation formatting 0.667 ($n = 12$), termination 0.100 ($n = 10$), test scheduling 0.333 ($n = 3$). In the mediation reading (Figure 3) $C_{Ha} \neq 0$ at 29/48: here the unmediated part of harness credit dominates. The registration fixed the gate threshold (non-screened mass ≥ 0.20) but under-specified the denominator and the treatment of 10 degenerate termination duals (continue $\rightarrow$ terminate makes $R_{HM} \equiv R_H$; addendum 29a): the gate opens under three accountings (0.50, 0.39, 0.21) and closes under the strictest (15/114 = 0.13; Table 6, Appendix E).

Both branches, exercised. The licensed training ran (pre-reg 32–33; four arms, 3 seeds, dose- and episode-matched; $\lambda^* = 0.2$ on 10 margin-verified tasks) and returned the declared outcome s3, attractor collapse: 8/9 runs collapse token-exactly onto summarize-always before any credit ordering becomes testable; a registered probe (pre-reg 34) locates the failure in the landscape (no λ yields even 3 tasks with attractor margin ≥ 0.10). On an 8B cell (pre-reg 39) the per-attractor power gate opened for the first time (min margin 0.166) and

the census closed it: the 8B is near-deterministic at the paired points (462/463 resamples identical), so no sampled-alternative estimand identifies $C_{HM} - C_M$, and the rule vetoed the training unpaid. The veto names the identifiability precondition: sampled-alternative credit is identifiable only where the model has entropy at the paired point, measurable in advance by a replay-free probe. Scope caveats ($n = 48$, 65 points excluded on entropy, mid-run amendments 29a/29b) are in Appendix E.

8 Threats to Validity and Discussion

Design of the pool, and the external test. The 36/37 comes from a pool designed so that a summarize decision destroys information the very next action needs, and at three quarters of the headline points that action is also the episode’s last write, where $C_{Ha} = 0$ is the expected reading; the measured content is the 21/23 at points with a later write, and the contrast with the 8B (2/25). The mediation fraction is a property of (decision type, model, pool), not a constant, and the designed-pool sample is small (16 unique pairs, 14 tasks). Pre-reg 42’s 48/49 rules out our pool as the cause but not the regime (43/49 saturated); one harness family, two models.

External validity and replication. Our tasks are designed, not sampled: the measurement needs control of destroyable-information position and deterministic fractional reward, jointly unavailable in public benchmarks. On MBPP+ and HumanEval+ (pre-regs 16/20/25) the floor replicates in every cell (3,824 nulls) and screening replicates unconditionally, while every pressure cell saturates away (pre-reg 37). Across models (Appendix D) the floor replicates everywhere; the 8B and the 4B-on-curved-pool break screening, so that claim re-scopes to the (model, harness, task) triple. Under the frozen V2 harness Mistral-7B reproduces the registered V2 outcome (pre-reg 38): no longer confined to one family (one cell).

Statistical and estimand caveats. Points within a task are not independent: the pressure-vs-slack contrast is 3 vs. 0 tasks (sign test $p = 0.125$; point level $p = 0.005$); only pre-registered outcomes (Table 9) are confirmatory, and saturation-conditioned rates are post-treatment. On the estimand (Appendix F): C_M and I share the sampled alternative; the single-draw a' is bracketed on both stacks (pre-regs 21, 26, 30); selection into measurement discards 53–73% of points, with composition pointing against the headline contrast; re-drawing a' flips $\sim 22\%$ of point verdicts while every aggregate survives (pre-reg 36).

The rule, v1.0. Prospective for any new cell: (i) measure the null floor (prefix caching off); trust no replay attribution while nonzero; (ii) run the entropy probe at candidate paired points: near-deterministic points are unidentifiable (pre-reg 39); (iii) run the census over all pivotal points, last-mover duals excluded; (iv) train on per-decision credit only if the census gate (≥ 0.20) and a per-attractor power gate both open — vetoes, not guarantees (Table 10); (v) bill the estimand that matches the trainable set: C_H (or the outcome-only gradient it is the expectation of) when only the harness trains, re-scaling the step for the exact signal’s sparsity; C_{Ha} or $C_{HM} - C_M$ only for joint training — controlled direct effects at mediator values a and a' , equal where $I_{\text{fact}} = 0$ — and only where the fourth-cell census shows $C_{Ha} \neq 0$.

9 Conclusion

A harness decision that changes the reward by nearly a full task changes it through the model’s very next action: at 36/37 reward-pivotal slack points — 48/49 on two external benchmarks, 21/23 where a later write could have intervened — the harness flip has zero effect once that action is held fixed. The fourth cell turns what three-arm replay (ours included until pre-reg 40) called “interaction” into the proximal mediated effect $PE = C_H - C_{Ha}$, $I = I_{\text{fact}} - PE$; mediation turns distal on the larger model, where its shield breaks (8B, 17/17), and is absent for decisions with a direct path by construction. The estimand follows the trainable set: C_H (or its replay-free expectation, outcome-only) when only the harness trains; direct effects, priced by a census first, for joint training (Appendix N).

Reproducibility Statement

All numbers derive from serialized artifacts (trajectories, replay rows, training logs) shipped with the code; make reproduce re-derives the null-floor, census, split, dose-response, margin, and selection statistics and fails on drift (Table 3). Serving premises: §4.1, Appendix L; estimands and controls: §3, Appendix F; ledger: Appendix J.

References

- Julius Adebayo, Justin Gilmer, Michael Muelly, Ian Goodfellow, Moritz Hardt, and Been Kim. Sanity checks for saliency maps. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2018.
- YanJun Chen, Yirong Sun, Hanlin Wang, Jinghan Wang, Xinming Zhang, Xiaoyu Shen, Wenjie Li, and Wei Zhang. Exact is easier: Credit assignment for cooperative LLM agents. *arXiv:2603.06859v2*, 2026a.
- Zhengyu Chen, Teng Xiao, Huaisheng Zhu, Yige Yuan, Luan Zhang, and Jingang Wang. Co-Harness: Co-evolving harnesses and model weights for LLM agents. *arXiv:2607.22688*, 2026b.
- Jakob Foerster, Gregory Farquhar, Triantafyllos Afouras, Nantas Nardelli, and Shimon Whiteson. Counterfactual multi-agent policy gradients. In *Proceedings of the AAAI Conference on Artificial Intelligence*, 2018.
- Jiarun Fu, Lizhong Ding, Sida Chen, Honglei Xin, Chunhui Zhang, Pengqi Li, Qiuning Wei, Ye Yuan, and Guoren Wang. CHILL-Harness: Counterfactual harness learning for efficient reasoning in long-horizon agents. *arXiv:2607.25825*, 2026.
- Anna Harutyunyan, Will Dabney, Thomas Mesnard, Mohammad Gheshlaghi Azar, Bilal Piot, Nicolas Heess, Hado van Hasselt, Greg Wayne, Satinder Singh, Doina Precup, and Rémi Munos. Hindsight credit assignment. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.
- Horace He and Thinking Machines Lab. Defeating nondeterminism in LLM inference. Thinking Machines Lab: Connectionism blog, 2025.
- Peter Henderson, Riashat Islam, Philip Bachman, Joelle Pineau, Doina Precup, and David Meger. Deep reinforcement learning that matters. In *AAAI Conference on Artificial Intelligence*, 2018.
- Raffi Khatchadourian. Replayable financial agents: A determinism-faithfulness assurance harness for tool-using LLM agents. *arXiv:2601.15322*, 2026.
- Hunter Lightman, Vineet Kosaraju, Yura Burda, Harri Edwards, Bowen Baker, Teddy Lee, Jan Leike, John Schulman, Ilya Sutskever, and Karl Cobbe. Let’s verify step by step. In *International Conference on Learning Representations (ICLR)*, 2024.
- Jiawei Liu, Chunqiu Steven Xia, Yuyao Wang, and Lingming Zhang. Is your code generated by chatGPT really correct? rigorous evaluation of large language models for code generation. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
- Mario Lucic, Karol Kurach, Marcin Michalski, Sylvain Gelly, and Olivier Bousquet. Are GANs created equal? A large-scale study. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2018.
- Haochen Luo, Yi Huang, Sichun Luo, Fengyuan Liu, Lei Li, Zefa Hu, Junlan Feng, and Qi Liu. Harness-aware self-evolving: Co-evolving model weights, harness, and task solutions. *arXiv:2607.03935*, 2026.
- Xufang Luo et al. Agent lightning: Train ANY AI agents with reinforcement learning. *arXiv:2508.03680*, 2025.

Judea Pearl. Direct and indirect effects. Proceedings of the Seventeenth Conference on Uncertainty in Artificial Intelligence (UAI), 2001.

Hans Reichenbach. The Direction of Time. University of California Press, 1956.

Rylan Schaeffer, Brando Miranda, and Sanmi Koyejo. Are emergent abilities of large language models a mirage? In Advances in Neural Information Processing Systems (NeurIPS), 2023.

Jaineet Shah. Causal agent replay: Counterfactual attribution for LLM-agent failures. arXiv:2606.08275, 2026.

David H. Wolpert and Kagan Tumer. Optimal payoff functions for members of collectives. Advances in Complex Systems, 4(2–3):265–279, 2001.

Haiyue Zhang. Credit without ground truth: Auditing step-level credit assignment in LLM agents against executed replay. arXiv:2608.19760, 2026.

Luan Zhang, Ruochen Zhou, Dandan Song, Zhengyu Chen, Yuhang Tian, Jun Yang, Huipeng Ma, Chenhao Li, Guangyuan Feng, et al. HarnessCompass: Guiding automatic harness evolution toward generalizable and effective agent harnesses. arXiv:2608.01918, 2026.

A Master Claim Table

Table 3 maps every claim of the paper to its evidence and epistemic status.

B Screening-Off, Formally

The measurements above are point identities, not expectations: under the premises of §4.1 (greedy decoding, fixed config, sequential serving, prefix caching off — with the resulting exactness measured, floor 0.0), the model, the environment, and the harness are all deterministic, so each replay branch is a single deterministic rollout. Fix a recorded trajectory with reward R_0 and a decision pair at index t : harness action h with alternative h' , model action a with sampled alternative a' . Writing $R_H = R(\text{do}(h'))$, $R_M = R(\text{do}(a'))$, $R_{HM} = R(\text{do}(h', a'))$ for the replayed rewards, the estimands are $C_H = R_0 - R_H$, $C_M = R_0 - R_M$, $C_{HM} = R_0 - R_{HM}$, and $I = C_{HM} - C_H - C_M$; throughout this section R is the test-fraction reward (not R_{eff} , whose token term accumulates over the differing prefixes), matching all measurements of §4.3. Screening-off at the point means $C_{HM} = C_M$.

Proposition 1 (Double-counting identity). At any point, $C_{HM} - C_M = C_H + I$. Consequently, where screening-off holds: (i) the marginal credit of the harness flip in the presence of the model flip is zero, $C_{HM} - C_M = 0$; (ii) the single-layer harness credit equals the negated interaction, $C_H = -I$; (iii) a trainer that bills the harness C_H at such a point assigns the total effect of the harness decision, all of it carried by the model’s response, while the signal $C_{HM} - C_M$ — the harness’s marginal contribution in the a' world — vanishes.

Proof. The first identity is the definition of I rearranged. Under $C_{HM} = C_M$, it gives $0 = C_H + I$; (i) and (iii) follow by definition. $\square$

Proposition 1 upgrades the empirical observation to a naming rule, with an honest scope note. C_H and $C_{HM} - C_M$ are the two orderings of a two-player marginal-contribution calculation, and which one a trainer should bill is a modeling choice tied to the training configuration: C_H is the advantage against a fixed model policy (the model replays its factual actions and adapts downstream), the correct estimand when only the harness trains; $C_{HM} - C_M$ is the harness’s marginal contribution in the world where the model deviates — a controlled direct effect with the mediator held at a' , as C_{H_a} is with the mediator held at a — the relevant wedge for joint training, where billing the harness for effects the model’s own update will absorb double-charges the same reward; a Shapley symmetrization would

bill the average, $C_H + I/2$ — computed over every recorded quartet, at screened points it bills exactly half of C_H , never zero. Our contribution is not to legislate the ordering but to measure the fan of admissible attributions — whose diameter is exactly I , usually the whole of C_H at screened points, obtainable for one extra replay — and, with the fourth cell, to show that the fan is a mediation fan: its diameter is $C_H - C_{Ha}$ up to I_{fact} (Remark 3). Training arm 3 used $C_{HM} - C_M$ and arm 2 used C_H ; they separated in Act 4 (§6) in the direction the mediation reading predicts.

Remark 1 (Additivity signature). The dual of screening is additivity: $I = 0$ with $C_H \neq 0$, i.e. $R_M - R_{HM} = C_H$ — the layers’ effects compose independently and single-layer harness credit is exactly correct (no double-counting). Which identity holds at a point is a property of the estimand of a' , not of the point alone: on the V2 stack, replacing the sampled alternative a' (drawn in the factual, information-rich state) with a'_s (the same sampler run in the counterfactual, post-summarization state) moves 13 of 13 non-screened points to exact additivity ($I_s = 0.0$, $C_H \neq 0$; pre-reg 30, Appendix F) — whereas the V1 de-screening under a'_s showed a re-injection signature instead. A census therefore licenses credit only under the estimand it was run with: the identity structure ($C_H = -I$ vs. $I = 0$) can flip with the estimand while the trajectory stays fixed.

Remark 2 (Mediation reading; why the joint arm is well-defined). Observationally the model action is a function of the harness decision, $a = f(h)$: change the context and the model’s next action changes with it. A recurring objection reads this as making the joint arm incoherent — “one world’s action inside another world’s context” — or redundant with C_H . The objection dissolves under the do-operator: the replay forces a' through the action queue, so in the joint arm the model is never queried and a' is a function of nothing — the intervention severs the edge $h \rightarrow a$, exactly as in causal mediation analysis (Pearl, 2001) with h as treatment, a as mediator, and R as outcome. In that vocabulary: C_H is the total effect of the harness flip (mediator free to respond); C_M is the mediator intervention under the factual treatment; C_{HM} is the joint intervention; and I is the mediation diagnostic. Screening-off ($C_{HM} = C_M$, hence $I = -C_H$) is total mediation — the entire harness effect flows through the model’s next action; additivity (Remark 1) is its absence. Note the arm the objection wants instead — re-query the model under the flipped context — is the C_H arm itself: building it as a “third arm” would be redundant by construction, which is precisely why the design forces a' . That the forced a' carries factual-world information into the counterfactual context is not a confound but the estimand choice the a' vs. a'_s contrast measures.

Remark 3 (Factorial cell and the status of I). Write $R_{Ha} = R(\text{do}(h', a))$ for the fourth cell of the 2×2 grid — the harness flip with the original next action forced — and $C_{Ha} = R_0 - R_{Ha}$ for the effect of the harness flip with the mediator held at its factual value. Because only the action at j is forced and later decisions respond live, C_{Ha} is the harness effect not carried by the immediate action; for context operations, which never touch the environment, it is carried by later model actions or is zero. Define the proximal mediated effect $\text{PE} = C_H - C_{Ha}$. The factorial interaction at the point is $I_{\text{fact}} = C_{HM} - C_{Ha} - C_M$, and

$$I = I_{\text{fact}} - \text{PE}.$$

The two coincide exactly when the model’s live response to h' carries no reward-relevant effect. The fourth arm (pre-regs 40–41, §4.4) measured $\text{PE} = C_H$ (i.e. $C_{Ha} = 0$) at 36/37 pivotal slack points and $I_{\text{fact}} = 0$ at 56/57. Hence at slack screened points $I = -\text{PE} = -C_H$ — Proposition 1(ii) is unchanged as algebra, but its reading is: single-layer C_H bills the harness a total effect that is entirely proximal mediation, because the factual action already re-injects the destroyed information. $C_{HM} - C_M$ remains the marginal contribution of the harness flip in the a' world; C_{Ha} is the estimand a joint trainer should prefer when billing the harness for effects the model’s factual action does not absorb.

Lemma 1 (Coalescence). Let $s_k = (c_k, e_k)$ denote the agent state at step $k \geq t$ — model-visible context and environment (sandbox) state — and consider the two branches $B_M = \text{do}(a')$ and $B_{HM} = \text{do}(h', a')$. If the agent is deterministic and there exists $k \geq t$ with $s_k^{B_M} = s_k^{B_{HM}}$, then the branches coincide from k onward; in particular $R_M = R_{HM}$ and screening-off holds exactly.

Proof. The one-step transition (model call, tool execution, harness policy) is a deterministic function of s_k . Equal states yield equal actions and equal successor states; induction to termination yields identical suffixes, hence identical final rewards. $\square$

Corollary 1 (Behavioral coalescence at the intervened turn). If both branches execute identical action sequences from turn t to termination, then $R_M = R_{HM}$. (Harness context operations do not modify the environment, so the environment states at turn t agree; identical action suffixes applied to equal environment states under a deterministic environment yield equal final environment states, and R is a function of the final environment state.)

The two empirical mechanisms of the anatomy are the two coalescence-inducing events we observe in practice; the lemma’s role is to guarantee that whenever coalescence occurs, the resulting equality is exact rather than approximate. Because reward equality does not certify coalescence (with reward resolution $1/n_{\text{tests}}$, collisions are possible), we verified the hypotheses directly in the recorded branch traces of all 75 screened points, comparing per-turn states (c_k, e_k) and per-turn behavior (action, observation), normalizing non-semantic identifiers that leak into test output (memory addresses, sandbox paths). The result, stated so that each count matches an enunciated argument: in 29/75 points the full state (c_k, e_k) becomes byte-identical within 1–4 turns of the intervention — Lemma 1’s hypothesis, verified; in 72/75 (a superset of the 29) the two branches execute identical action suffixes from the intervened turn itself — Corollary 1 applies and entails $R_M = R_{HM}$. The formal arguments therefore explain 72 of the 75 measured screening events. The 3 residuals: two (l_log_parser idx 10, in two configurations) are exactly the reward-collision case — $R_M = R_{HM}$ holds without trace coalescence, so the screening statistic counts them but the theory does not explain them — and one (l_payroll_pipeline idx 7 under g900, saturated at $R = 1.0$) coalesces behaviorally only 3 turns after the intervention, which is consistent with screening but not covered by Corollary 1; we declare all three rather than absorb them into the explained count. No suffix violated determinism: whenever states matched at some turn, the branches were identical from that turn to termination, corroborating the premises at trace level. Re-triggering (22/52): both branches executed the same a' , so environment states already agree; the live harness in B_M summarizes at step $t+1$, and context coalescence follows. Re-injection (23/24 annotated): a' (typically a write_file carrying the destroyed constants) equalizes the environment component, and the residual context difference is confined to content the greedy continuation never conditions on. The lemma thus explains a striking feature of the data: measured screening-off consists of exact zeros, not small values — determinism converts coalescence into equality, the measured 0.0 floor makes equality observable, and the coalescence itself is now measured rather than presumed.

Remark 4 (Pivotality and the competence window). Call a decision point reward-pivotal if at least one of the three interventions changes the reward. Non-pivotal points satisfy $C_H = C_M = C_{HM} = I = 0$ identically: nothing is detectable there, by construction rather than by noise. When the agent fails a task outright and no intervention — single or joint — rescues it, every point on that trajectory is non-pivotal. This is the formal content of the competence window: reward saturation does not merely add variance — it removes pivotal points, which is why the Qwen3-1.7B pressure regime (11/12 saturated) had essentially no power and why the mt4 break rate collapses.

Corollary 2 (What single-layer training optimizes). In a regime where screening-off holds at all measured points (our slack regime, 57/57), the per-decision signal of a single-layer harness trainer is $-I = C_H$ at every shielded point, and where in addition $C_{Ha} = 0$ (36/37 pivotal slack points) it equals PE: the total effect of the harness decision, all of it carried by the model’s next action (at the one screened point with $C_{Ha} \neq 0$, l_log_parser, $-I = C_H = -0.154$ while $\text{PE} = -0.231$). This is the correct signal when the model policy is fixed and the wrong one when the model trains too, since the model’s own update can move PE. By Proposition 1 the two candidate signals C_H and $C_{HM} - C_M$ differ by exactly I at every point where $I \neq 0$, regardless of regime — at the broken point l_vending_machine, $C_H = -0.09$ while $C_{HM} - C_M = +0.09$, an inverted sign.

Screening-off has a direct budget consequence for any attempt to train on replay credit — one that is decidable before training begins. We state it here as the operational centerpiece of the paper; §6 then tests it the hard way.

Proposition 2 (Compute dominance: replay credit under screening). Fix a task pool, the determinism premises of §4.1, and a training budget matched in total LLM calls. Let σ be the fraction of reward-pivotal harness points at which screening-off holds exactly, and consider a trainer that bills the harness the corrected signal $C_{HM} - C_M$, paying $r \geq 2$ replays per billed point out of the shared budget. Then: (i) every screened point contributes a billed signal of exactly zero (Proposition 1(i)) at strictly positive replay cost; (ii) the total billed per-decision signal is confined to the $(1 - \sigma)$ non-screened pivotal mass and bounded by the effect sizes $|C_{HM} - C_M|$ there — so any training value that flows through the billed signal is bounded by (non-screened mass $\times$ effect size) minus the replay tax; (iii) if $\sigma = 1$ on the measured pool, the consequence depends on the update rule, and both cases are dominated: a trainer that bills only the corrected credit receives an identically zero gradient and reduces to the untrained policy — which is exactly what Act 4 measures (the corrected arm lands on the untrained control, 0.392) — while a trainer that mixes an outcome term with the (zero) credit term reduces to outcome-only policy gradient computed from a strictly smaller episode budget.

Proof. (i) is Proposition 1(i) plus the accounting of dose-matching. (ii) sums (i) over points: screened points contribute zero billed signal, non-screened points at most their effect size, and every replay is paid from the shared call budget. (iii) case one: with the billed credit identically zero, every per-decision update vanishes and the policy stays at initialization; case two: with the credit term zero, the update rule coincides with outcome-only on the episodes it can still afford. $\square$

The bound in (ii) prices the billed signal, not every conceivable channel of value: a correctly-zero credit could in principle still help as a variance-reduction baseline against the noisy outcome return assigned to the same points. We state this channel explicitly rather than bound it; on our stack Act 4 shows no such benefit (the corrected arm does not outperform outcome-only anywhere), but the proposition does not exclude it elsewhere.

Corollary 3 (The census is a decision procedure). Both quantities in the bound — σ and the non-screened effect mass — are measurable before any training by a screening census whose cost is a small multiple of baseline collection. Whether cross-layer replay credit can pay for itself on a given stack is therefore decidable in advance: pre-register a gate on the census and run credit training only if it opens. With the fourth cell (Remark 3) the census has a mediation reading: σ is the fraction of pivotal harness points whose effect is fully carried by the next model action ($C_{Ha} = 0$), and the non-screened mass is the harness effect that survives fixing that action — the only part a joint trainer can bill to the harness without the model’s update moving it.

Three scope notes keep the proposition honest. First, the dominance in (iii) is a property of the measured pool, not of the method class: pools with substantial non-screened pivotal mass may repay credit — the proposition prices replay credit, it does not prohibit it. Second, the tax term attaches to the replay estimator under dose-matching, not to the signal itself: an amortized critic fit offline to census data moves the replay cost out of the training loop and changes the accounting — measurement remains the right place for replays, and the census is exactly the dataset such a critic needs. Third, screening annihilates the harness-side signal only: C_M is untouched, so model-side credit is priced by a different budget line. Under this reading, the corrected arm’s null in Act 4 is not an optimization pathology but the correct behavior of an exact estimator of a quantity that is truly zero; the C_H arm’s loss is a separate, optimization-side fact (§6).

We emphasize the epistemic status of each piece. Proposition 1 is elementary algebra — its value is not mathematical depth but the naming rule it licenses. Lemma 1 holds given the determinism premises, and both the premises (§4.1) and the lemma’s hypothesis (this section) are measured rather than assumed. Whether coalescence occurs at a given point, which mechanism induces it, and how often screening-off holds at all are empirical questions that the rest of this section answered.

B.1 Control tasks and the `c_temp_label` exception

On control tasks, where summaries destroy nothing by construction, the method returns $C_H = 0$ in 11/12 measured points — the construct behaves. The exception is informative: `c_temp_label` showed $C = +0.22$ at turn 0 despite all task-critical information surviving the summary. Audit of the replay pair shows the effect is behavioral, not informational: the summarized context changes the model’s subsequent formatting behavior, which a strict test then penalizes. C_H measures total causal effect through any pathway; the controls certify the informational pathway is clean, and the exception demonstrates the behavioral pathway exists and is detectable.

C Shield Anatomy and Robustness

Table 4 lists every screening-off count in the paper with its endpoint.

Anatomy of the shield. The sets in play: across the three slack configurations and the first pressure configuration we measured 78 interaction points (57 slack + 21 at mt6); 75/78 show exact screening-off ($C_{HM} = C_M$) — the 3 exceptions are the mt6 pressure breaks. We report this estimation-first, on the falsifiable subpopulation: by Remark 4, reward-non-pivotal points satisfy $C_{HM} = C_M$ by construction, so the informative unit is the 56 reward-pivotal points. There the screened fraction is $53/56 = 0.946$, CI95 [0.87, 1.00] by task-clustered bootstrap over the 16 pivotal tasks; collapsing further to unique task×point pairs (removing cross-configuration pseudo-replication) gives 19/22, and 13 of 16 pivotal tasks screen at every measured point. Binomial sign tests against a 50/50 null give $P = 4.1 \times 10^{-13}$ (instances) and $P = 4.3 \times 10^{-4}$ (unique pairs), but we do not rest the claim on them: no substantive model of the data-generating process fixes the non-screening probability at 0.5, so these p -values certify only that the pattern is not coin-flip noise. The inference we retain is the clustered estimate; a bundled script (`estatisticas_pivotaux.py`) re-derives every number in this paragraph from the raw artifacts and fails on drift. One further split the timeline forces on us: the screening-off hypothesis was formulated after seeing the first slack census (g600, whose artifacts predate the registration commit by 34 minutes), so g600 is discovery data. Splitting accordingly: discovery (g600) screens 14/14 pivotal points; the confirmation set (g450/g900/mt6; g450 was in flight at registration, g900 and mt6 strictly subsequent) screens 39/42; restricting to the two strictly post-registration configurations (g900+mt6) gives 27/30. The estimate does not depend on the discovery cell.

Robustness to the alternative draw (pre-registration 21). The sharpest objection to every number above is that C_M , $C(HM)$, and hence screening are relative to a single sampled a' per point. We pre-registered outcomes (stable ≥ 0.90 / partial / fragile < 0.75 , over informative slack draws) and re-sampled a' at all 56 pivotal instances under three disjoint seed schedules (3001–3008, 4001–4008, 5001–5008; same procedure, temperature 0.8), yielding 152 new draws (16 no-alternative outcomes; 304 extra replays), of which 81% differ from the published a' . The pre-declared stable outcome was met: 99/101 informative slack draws screen exactly (rate 0.980, CI95 task-clustered [0.944, 1.00]); restricted to genuinely new alternatives, 79/81. The two non-screened draws are themselves informative: one is a single-draw excursion at the already-declared reward-collision residual (`l_log_parser` idx 10¹), and one is a genuine synergy break at `invoice_pricing` under one of its three draws ($I = +1.57$; the other two draws screen) — so screening at a point is a dominant-but-not-sure property of the alternative distribution, not a logical constant. Symmetrically, the three published mt6 break points break again under every redraw (9/9): breaks are properties of the decision point, not of the draw. Sixteen draws found no alternative (the selection channel of §8, rate consistent across schedules: 6/4/6). Among the screened points, 52 are non-trivially shielded: the harness flip alone has a real effect ($C_H \neq 0$) that the joint flip erases — these are exactly the double-counting cases. Of the 52 shielded points, 24 carry constant-dictionary annotations that make the re-injection check possible (the remaining 28

¹Point labels throughout are checkpoint indices (`cp_index`); raw artifacts also carry the 1-based tool-call index ($= \text{cp_index} + 1$), so this point appears as index 11 in `results.jsonl`.

are unannotated V2 points, which we declare rather than extrapolate). Among the annotated: 23/24 show structural re-injection — the sampled alternative action a' carries the destroyed information across the cut, a property of the estimand (since a' is sampled from the model conditioned on the un-summarized state); 9 also show the second mechanism and 1 shows neither. The second mechanism is checkable on all 52 shielded points: in 22/52 the live downstream harness re-triggers summarization anyway, equalizing the branches. Section B makes this division precise: a coalescence lemma and a one-line corollary guarantee that whenever the two branches coalesce, the resulting equality is exact — and we verify the occurrence directly in the recorded traces (the arguments explain 72/75 screened points; the 3 residuals, including 2 reward collisions, are declared). We also found genuine non-saturated synergy where the shield breaks (1_vending_machine: $C_H = C_M = -0.09$, $C_{HM} = 0$, so $I = +0.18$: either flip alone hurts, together they cancel).

D Replication Across Models, Pools, and Environments

Table 5 synthesizes raw screening-off break counts by (model, pool, regime); the paragraphs below give each cell’s record.

The cross-family attempt (pre-reg 35). A census mirroring pre-reg 15 on a non-Qwen model was registered with a protocol-adherence smoke gate: 3 episodes, action-parse rate ≥ 0.80 , no screening analysis before passing. Three ungated candidates failed it in sequence (addenda 35a–c, each registered before contact with the next model): Mistral-7B-Instruct-v0.3 at 0.57 — after a role-merging serving shim its strict-alternation chat template required — DeepSeek-Coder-6.7B at 0.33, and Phi-3.5-mini at 0.43. A raw-output probe confirmed the failure mode is not our parser: the models answer in prose plus fenced code, never emitting the action object. The declared stopping rule (no fourth candidate, no harness relaxation) fired: under a frozen harness, adherence to the action format — the precondition of measurement — is itself family-dependent — but pre-reg 38 reclassifies the boundary: the barrier is the V1 JSON protocol, not the family.

Cross-family under the frozen V2 harness (pre-reg 38). The V2 stack uses a plain-text action protocol adopted for reasons independent of and prior to any cross-family design (addendum 28a). Re-registering the cross-family question against the frozen V2 harness — same smoke gate, a fresh instrument-validity gate (greedy null-flip floor, exact ≥ 0.95), candidate order and stopping rule declared, no per-model adjustment — Mistral-7B-Instruct-v0.3 passes the smoke at 1.00 (66 parses; 0.57 under the V1 JSON envelope) and the full census chain ran on the first candidate. The instrument transports: 48/48 nulls and 48/48 greedy null-flip replays exact. The registered outcome reproduces the Qwen cell at the census and gate levels: census verdict s3 (43 valid points; observation 0.62 $n=21$, context 0.83 $n=6$, termination 0.13 $n=15$), the gate opens under three accountings and closes under the strictest (primary measured-without-duals 9/28 = 0.32 vs. Qwen 15/38 = 0.39 under the matched accounting; pivotal-without-duals 0.18 vs. 0.13, both closing), and the a'_s coupling lands in the same declared bucket (3/6 vs. Qwen 0.381; $n=6$, CI95 spanning all declared buckets — support, not an estimate). Termination is the least screened type in both families, but that agreement is largely by construction: both cells’ termination points are dominated by analytically degenerate last-mover duals (addendum 29a); the informative transport is in context and observation. Selection composition differs and is reported: the Mistral sampler found an a' at every census point (0 excluded, vs. 65/114 for Qwen), so the measured populations are selected differently — cross-family numerical rate comparisons are therefore confounded by selection, and what transports is the registered outcome level (s3, the gate pattern, the coupling bucket), not per-type point estimates. One cell, one harness, one environment: the finding is no longer confined to a single model family — not “generalizes across families.”

Floor and unconditional replications. The floor replicates everywhere (pre-regs 14, 15, 16, all registered before running): the 8B contributes 810 dedicated and 234 attached null replays and MBPP+ contributes 1,294, all exact. Screening-off replicates unconditionally on Qwen3-1.7B (0 raw breaks in 23 points) and on MBPP+ (66/66 raw; 11/11 pivotal).

The 8B breaks the shield. On Qwen3-8B it does not: raw break rates are flat across regimes (5/28 slack, 6/34 mt6, 7/35 mt4), so the slack/pressure contrast established on Qwen3-4B does not transfer to the 8B, and we report this as the headline of the replication rather than conditioning it away. Saturation-based exclusion would be invalid here: clipping R at $\{0, 1\}$ can only shrink $|C_{HM} - C_M|$, so saturation can mask nulls but cannot fabricate breaks — a saturated break of 0.9 is a real break. The 8B slack breaks have a sharp common structure ($C_H = 1.0$, $C_M = 0.0$, $C_{HM} \in [0.8, 0.9]$, hence $I \in [-0.2, -0.1]$): the harness flip alone is catastrophic, the model flip alone is null, and the joint flip tracks the harness — the shield inverts. The mechanism is the one the anatomy of §4.3 identified: on the same tasks the 4B’s sampled a' re-injects all destroyed critical constants (5/5, 5/5, 6/6 on the shielded points), while the 8B’s a' re-injects none (0/5, 0/6, 0/8, 0/5; matching by constant-literal substring in the serialized a') — its solutions are structured so that the alternative action no longer carries the constants. The mechanism is verifiable on 4 of the 5 slack breaks (the fifth, `invoice_pricing`, has no annotated constants), and the breaks recur: all 5 slack break tasks break again, with identical per-task structure, in both pressure configurations (6/34 at mt6, 7/35 at mt4). Whether screening-off holds is therefore itself competence-dependent: at that stage of the design the shield appeared to be a property of the model–harness pair — a claim the symmetric control below narrows further to the (model, harness, task) triple (the closest precedent, model-dependent counterfactual measurability (Zhang, 2026), concerns whether ground truth can be estimated at all; here the recovered causal structure itself inverts). We say competence, not scale: 4B and 8B differ in post-training as well as size, and the claim is anchored in the measured re-injection contrast on the original pool (which, as reported below, does not generalize as a mechanism), not in parameter count. Under pressure the 8B also shows a genuine non-saturated synergy break at `l_door_controller` ($C_{HM} = 0.23$ vs. $C_M = -0.46$, $I = +0.38$). Saturation still dominates both auxiliary models in opposite directions (11/12 of the 1.7B’s pressure points, from weakness; 90/97 of the 8B’s points, from strength — and the slack regime leaves the 8B a single non-saturated point, so the conditioned slack contrast has essentially no power and we do not claim it).

The 8B’s fourth cell (pre-reg 42, secondary). Replaying $R(h', a)$ at all 97 8B census points (g600/mt4/mt6; null gate 6/6 exact on the new hardware) splits the cell along the shield: at the 57 pivotal screened points $C_{Ha} = 0$ in 51 (0.895; 21 unique pairs, 18 of them in every configuration), whereas at the 17 pivotal non-screened points — the five cp0 break tasks above in all three configurations plus `l_door_controller` — $C_{Ha} \neq 0$ in 17/17, with $I_{\text{fact}} = 0$ exactly at 15 of them ($C_{HM} = C_{Ha}$, $C_M = 0$: the harness effect is additive and unmediated, neither the original action nor the sampled a' re-injects the constants). The six screened breaks come from three tasks (`l_log_parser`, `api_router`, and one `l_door_controller` point); in five of them the point is screened only because both a' arms fail the task outright ($C_M = C_{HM} = 1.0$), and there the fourth cell shows $C_{Ha} \approx C_H$: screening by failure saturation is not mediation. Non-pivotal points are exact at 23/23. The 8B is thus the cell where the census’s two readings coincide point by point: non-screened mass is direct-effect mass.

Curated pool (pre-registrations 17–18). The remaining step — the break rate on a pool curated to the 8B’s competence — was then run (pre-registrations 17 and 18): a first hardened pool failed its pre-registered curation rule (17/24 tasks solved at 1.0; reported as failure), and a second pool targeted at the fractional-reward profile passed (22 of 44 tasks in the (0.05, 0.95) window; 792/792 nulls exact after the ASLR fix of §4.1). In-window, the 8B’s raw screening breaks replicate and stay flat across regimes — 3/27 slack, 3/31 pressure — including a non-saturated slack break (`x_hours_bank`: $C_H = +0.25$, $C_M = -0.17$, $C_{HM} = 0.00$, $I = -0.08$), which closes the clipping objection: breaks persist where saturation cannot manufacture them. The pool-matched competence contrast is the 8B replication’s (pre-reg 15; same 30 tasks: 4B 0/57 raw vs. 8B 5/28); within the curated pool the regime contrast seen on the 4B does not exist on the 8B even inside its competence window. Two of the three break points recur across both regimes with identical structure. Two pre-registered sub-hypotheses failed: the binary re-injection mechanism does not generalize (at active slack points the curated-pool a' never re-injects constants, 0/15, yet only 3/15 break; under pressure 2 of 3 breaks occur with re-injection; one-sided Fisher $p = 1.0$ and $p = 0.94$ — the slack table is degenerate, no exposed points, so only the pressure test is informative, and it points the wrong way), and the directional prediction (pressure $\geq$

slack) failed (9.7% vs. 11.1%). All curated-pool break counts are descriptive: with 3 break tasks per regime we make no clustered inferential claim, and we make no cross-pool p -value comparison against the 4B’s 0/57.

Symmetric control (pre-registration 19) and the triple claim. Two design threats were named at this point — the curated comparison was cross-pool (the 4B’s 0/57 comes from the original pool — the paired contrast on identical tasks is pre-reg 15’s 0/57 vs. 5/28); and the curation window is operationalized by task family (the second pool was designed around the profile that survived curation), so in-window break rate could be a property of the family rather than of competence — and the pre-registered symmetric control (pre-reg 19: the 4B on the same 22 curated tasks, outcomes declared in advance) resolved both, against the strong form of our own claim. The 4B also breaks raw screening on the curated pool: 4/16 slack points (3 non-saturated, including a synergy break at `h_turnstile_fsm` with $I = +0.77$ and `h_hotel_folio` at $I = -1.42$) and 2/23 pressure points; its floor held (396/396 nulls exact), and the pivotality-based prediction that the 4B would sit saturated-in-failure on this pool failed (1/16 and 3/23 saturated — reported as a failed pre-registered sub-hypothesis). The competence claim therefore re-scopes: the shield is a property of the (model, harness, task) triple, not of the model-harness pair alone.

What survives every cell. The pool-matched model contrast on identical tasks (0/57 vs. 5/28), the majority of points screened in every configuration (worst cell 12/16), the regime contrast existing only for the 4B on its original pool, and a majority pattern of break-point stability: break sets are disjoint across models at the point level (the two models share one break task, `h_hotel_folio`, at different turns), and within a model most break points recur across regimes — in the 8B, 2 of 3 slack break tasks recur numerically identically; in the 4B, 2 of 3 break tasks recur, one numerically identically (`h_hotel_folio`), one with the same sign structure but different magnitudes (`h_turnstile_fsm`: $I = +0.77$ slack vs. $+0.31$ pressure), and `h_telco_roaming` does not recur. A further reading — breaks concentrate where the pool sits near the model’s competence boundary (for the 4B, the original pool under pressure and the hardened pool in both regimes; for the 8B, the hardened pool in both regimes) — is descriptive and not pre-registered: the pre-declared interpretation of outcome c2 was “a property of the task family, not of competence,” and what favors the triple reading over the pure-family reading is the point-level disjunction of break sets between models on the identical pool. The break structure is also heterogeneous — the original-pool 8B slack signature ($C_H = 1.0$, $C_M = 0.0$, $C_{HM} \approx 0.9$) does not recur — and a descriptive trace anatomy of all 8B and 4B curated-pool breaks (0 new rollouts) finds one uniform base, three recurring patterns, and a model asymmetry (restore-direction flips appear only in the 4B); Appendix M reports it in full, including two audit details showing that recurring break points are numerically identical across regimes while I magnitudes are baseline-relative. The robust claim is therefore narrower than the mechanism story: whether screening-off holds is a property of the (model, harness, task) triple, and re-injection of destroyed information, which fully explained the annotated 4B/8B contrast on the original pool (pre-reg 15), is neither necessary nor sufficient for the shield in a curated pool.

External environments in full (pre-regs 16, 20, 25). We therefore ran the missing experiment in reduced form (pre-reg 16, registered before running): a multi-turn MBPP+ adaptation — 60 tasks derived mechanically from MBPP+ (Liu et al., 2023) with fractional reward (6–10 test functions per task, canonical solutions validated in our sandbox), one slack and one pressure configuration on Qwen3-4B. The floor replicates exactly (1,294 null replays, all exact) and screening-off replicates at 66/66 interaction points, of which 11 are reward-pivotal (11/11 screened) — the other 55 screen by construction (Remark 4), so the informative content of the replication is the pivotal 11/11, not the 66/66 — including a non-saturated point with $I = -0.4$, a live double-counting case in an external benchmark. What the adaptation cannot yet test is the pressure-regime break rate: 60/66 points are reward-saturated (Qwen3-4B is too strong for MBPP+), leaving 6 non-saturated points and no breaks — inconclusive at that power, the same saturation boundary seen at 1.7B and 8B. A pre-registered second attempt (pre-reg 20: the 1.7B on the same MBPP+ pool, declared outcomes) landed on the same boundary from the same side: floor exact (1,080/1,080), screening replicates raw at 44/44 points across both regimes (24/24 reward-pivotal) — a

second (model, environment) cell with unconditional screening in the external benchmark — but 20/22 points per configuration are saturated (the 1.7B still solves 34/60 MBPP+ tasks at 1.0) and the pressure budget never binds on MBPP+’s short trajectories, so the pressure cell of environment B remains empty after two attempts, and we declare the cell empty. A third environment (pre-registered as pre-reg 25 with the same frozen mechanical derivation, from HumanEval+ (Liu et al., 2023): 60 tasks, 164 processed, zero discarded) reproduces the pattern a third time: floor exact (1,294/1,294 nulls) and screening at 76/76 interaction points, of which 42 are reward-pivotal (42/42 screened), with the pressure cell again saturated away (2 non-saturated points) — declared inconclusive by the pre-specified power gate.

E V2 Census: Gate Accounting and Scope

The V2 registration (pre-reg 29) fixed the gate threshold (non-screened mass ≥ 0.20) but left the denominator (measured vs. all pivotal points) and the treatment of the 10 degenerate termination points under-specified; the verdict turns on both (Table 6). We report all four accountings.

Last-mover screening. Terminal flips (continue→terminate) make the joint arm analytically degenerate ($R_{HM} \equiv R_H$ by determinism; addendum 29a). These 10 termination points mirror the V1 finding with the layers swapped: screening-off is a fact about which layer moves last on the channel that reaches the reward — and, in the fourth cell, the only type with $C_{Ha} \neq 0$ in every point (§4.4).

The 8B false negative. An 8B follow-up (Appendix N) turned the under-specification into a false negative: the per-task power criterion vetoes (0/24) even where the conditional target dominates every fixed policy in the mean (+0.16). The registered per-attractor correction (pre-reg 39) opens the power gate ($\lambda^* = 0.1$, min margin 0.166); the census then closes it twice — the 8B is near-deterministic at the paired points (462/463 resamples identical) and recomputing the census in $R_{\text{eff}}(\lambda^*)$ over the 92 paid flips returns the identical pivotal set (token-only effects ≤ 0.033).

Scope. Three caveats bound the verdict. Power and selection: $n = 48$ with 65 points excluded for want of an a' makes per-type rates wide, and the exclusion selects on model entropy — at excluded points the model is near-deterministic (the correction is vacuous there by construction), and worst-case imputation of all 66 excluded points cannot close the measured-accounting gate ($24/114 = 0.21 \geq 0.20$) — a bound covering the with-duals accountings only: without the ten duals, imputation spans 0.13–0.71, straddling the threshold; test scheduling ($n = 3$) supports no claim; restricting to types with $n \geq 10$ leaves s3 unchanged (context 14/23, observation 8/12). Estimand: pre-registered controls (pre-reg 30, Appendix F) discipline the verdict: it survives both a' re-draw schedules, and under the information-poor a'_s the de-screened points turn exactly additive ($I = 0$) — the V1-closed / V2-open contrast compares verdicts under estimands of differing completeness. Addenda: two amendments (29a, 29b) were registered mid-run (before analysis, after data contact); the verdict inherits that weaker label where they bind.

F Estimand and Infrastructure Caveats

Multiplicity. This paper reports on the order of a dozen p -values across regimes, strata, and clustering levels. Only the pre-registered outcomes (Table 9) are confirmatory. A post-hoc Benjamini–Hochberg and Holm correction over the six screening hypothesis tests confirms the caution: only the point-level $p = 0.005$ survives either correction (Holm-adjusted 0.03); $p = 0.037$ does not (Holm-adjusted 0.19). The horizon family (pre-reg 41c, six populations: V1 slack, V1 pressure, V2 all non-termination types, V2 context, V2 observation, external 4B; one-sided Mann–Whitney on H) was re-analysed under pre-reg 43 with Holm and with a task-level permutation test (20,000 permutations of the task label “has any $C_{Ha} \neq 0$ ”, statistic = difference in mean per-task median H): raw $p = 0.026 / 0.0008 / 0.031 / 0.011$

/ 0.43 / 0.037; Holm-adjusted 0.10 / 0.005 / 0.10 / 0.056 / 0.43 / 0.10; task-permutation 0.073 / 0.40 / 0.27 / 0.099 / 0.89 / 0.24. Only V1 pressure survives Holm, and nothing survives clustering; the horizon contrast is reported as a uniform descriptive direction, not a finding. The claims we retain rest on replicated structure — exact screening at pivotal points, cross-configuration recurrence of break points, the pool-matched model contrast, and the stratified fourth cell — rather than on any single threshold crossing.

Saturation is endogenous, and conditioning on it is post-treatment. Reward resolution is set by test count, so saturation is partly a design artifact. Worse for inference: saturation is defined on the rewards of the replayed arms, i.e. on outcomes of the intervention itself, so every saturation-conditioned rate in this paper conditions on a post-treatment variable and carries selection bias of unknown sign. We therefore use conditioned contrasts descriptively only, never confirmatorily, and we describe saturation as a correlate of the competence window, not a mediator. The inverted-U at mt4 is only partially accounted for (conditioned rate $0.045 \rightarrow 0.143$; the ordering $\text{mt8} > \text{mt6} > \text{mt4}$ persists), so the window remains a partially supported hypothesis, discussed but not claimed as a contribution.

Estimator coupling. C_M and I share the sampled alternative action a' : they are not independent evidence, and the structural component of screening-off (§4.3) is partly a property of the estimand — a' sampled from the un-summarized state can re-inject information. We quantified this (23/24 annotated shielded points; 28 unannotated points declared) rather than hiding it; an estimand with a' sampled from the summarized state would measure a different, also legitimate, quantity. Two further limitations of the estimand: a' is a single sample per point (fixed seed schedule, §3) in the census; the pre-registered re-sampling study (§4.3) measures sensitivity to that draw directly and finds the slack result stable (99/101) and the pressure breaks draw-invariant (9/9). What remains unmeasured is coverage of the alternative distribution beyond $k = 3$ redraws at temperature 0.8. The alternative estimand — a'_s conditioned on the summarized state, so that re-injection is impossible by construction — we ran as a registered follow-up (declared outcomes; all 44 keep→summarize pivotal instances, one seed schedule, 2 replays each, zero sampling failures, all 44 a'_s distinct from the published a'). The result splits cleanly: screening under a'_s holds at 24/31 informative slack points (0.774, task-clustered CI95 [0.69, 0.88] — the pre-declared “partial” bin, down from 53/56 under a'), and all seven de-screened points have the same signature: $C_M(a'_s) = 0$ and $C_{HM}(a'_s) \approx C_H$ — exactly what turning off the re-injection channel predicts. The two estimands therefore bracket the phenomenon: with an information-complete alternative the model absorbs essentially everything (53/56); with an information-poor alternative the harness flip’s damage reappears in the joint branch at about a quarter of the points. Screening is real in both estimands but its completeness is partly a property of the alternative’s information access — the headline claim is conditional on the estimand, and we scope it so. Under pressure the picture inverts symmetrically: both published mt6 break points in this direction (l_log_parser idx 7, api_router idx 0) screen under a'_s — so break-ness, too, is a property of the (point, alternative) pair rather than of the point alone, consistent with the stochastic cell where the weakest break dissolved under re-sampling. (The third published break is a restore-direction flip, outside this estimand’s population by construction.)

V2 estimand controls (pre-reg 30). The same two controls, run on the V2 census’s 48 measured points. Re-sampling (two disjoint seed schedules at each point’s census temperature): the plain-text sampler is near-deterministic even re-seeded (56/96 draws return the identical a' ; 17 find none), and on the 23 informative draws the point-level classification is draw-sensitive — stability $17/23 = 0.739$, with flips in both directions across three decision types. The aggregate verdict is not: substituting each schedule’s draws and recomputing, the census outcome remains s3 and the primary gate stays open under both schedules (0.474 and 0.368 vs. 0.395 original; threshold 0.20). A dedicated distributional characterization (pre-reg 36; five further disjoint schedules, up to eight draws per point) confirms and sharpens this: the a' distribution is concentrated (median one distinct canonical a' per point, max six; 139/240 re-draws return the census a'), verdicts are unanimous at $36/46 = 0.78$ of points with flips confined to context-policy (7/21) and test-schedule (2/3) types — termination is immune (0/10) — and the headline is draw-invariant: s3 and the open gate hold under all five schedules (non-screened fraction 0.42–0.50) and under per-point majority vote

(0.42). Summarized-state a'_s (23 keep→summarize context points, one schedule): screening drops to $8/21 = 0.381$ — and all 13 non-screened points (9 genuinely de-screened) share a signature different from V1’s: $I(a'_s) = 0$ exactly with $C_H \neq 0$, i.e. the layers become additive — the one regime where single-layer credit is correct. Which identity holds at a point — screening ($C_H = -I$), additivity ($I = 0$), or neither — is itself a function of the estimand, which sharpens the prescription: run the census under the estimand you intend to bill. Selection composition, V2: the 66 pivotal points excluded for lack of an a' have flip effects comparable to the measured 48 (median $|\Delta R|$ 0.167 vs. 0.182) but sit earlier (median turn 2 vs. 7) and in smaller contexts (919 vs. 2,099 tokens), with termination and observation over-represented and both retry points excluded — the census over-represents late, large-context states where the sampler has entropy, a direction-neutral but real selection.

Selection into measurement. A point enters the interaction census only if the sampler finds a parseable $a' \neq a$ within the 8-seed schedule; across configurations 53–73% of candidate points are discarded this way. The measured population is therefore selected on the model’s own output entropy at that state — a post-decision variable plausibly correlated with both competence and the information content of the context — and every break or screening rate is conditional on “an alternative exists.” For the headline cross-model contrasts the discard rates are close enough that differential selection is an implausible driver: on the pool-matched contrast (identical 30 tasks) the 4B discards 66–73% across its slack configurations versus 64% for the 8B at the same budget, a ≤ 9 -point selection gap against a 0% vs. 18% raw break-rate gap; on the curated pool the rates are 57–70% (4B) versus 53–59% (8B). We measured the composition directly (analyse_selecao.py, from the per-point sampling logs joined to the baseline trajectories): selection is real and directional — in both models the retained points sit earlier in the trajectory (pool-matched relative position, found vs. discarded: 0.33 vs. 0.73 for the 4B, 0.21 vs. 0.79 for the 8B; task-clustered bootstrap CIs exclude zero) and carry smaller contexts. But the direction disfavors the headline contrast rather than explaining it: the 8B’s retained points are shallower than the 4B’s (mean turn 0.65 vs. 1.55, context 395 vs. 433 tokens, $d = 0.31$ – 0.54), so if depth promotes breaks, composition biases the 8B toward fewer measured breaks — and it still breaks at 18% against the 4B’s 0%. What selection does forbid is any claim about deep-context interaction: no census measures points the sampler cannot reach, and the discarded majority is precisely the late-trajectory, large-context population.

Serving premises. The exact floor holds under a fully specified stack: vLLM 0.8.5.post1 (V1 engine, pinned; PyTorch 2.6.0+cu124, CUDA 12.4, driver 580.173.02, Python 3.11.14, full transitive closure pinned in the repository’s uv.lock) serving one model per experiment on a single NVIDIA RTX 4090; dtype auto (bfloat16); CUDA graphs enabled (no enforce_eager); default chunked prefill (max_num_batched_tokens = 2048); automatic prefix caching explicitly disabled; GPU memory utilization 0.85 (0.92 for the 8B); strictly sequential requests against a single instance; greedy decoding (temperature 0, single seed) for every episode and replay rollout; and the fixed a' seed schedule of §3. The floor is a measured property of this stack, not a theorem; any deviation (batching, caching, config drift, a version bump) voids it and must be re-measured. Pre-reg 42 ran on different hardware — one NVIDIA H100 80GB per job under Slurm, driver 580.159.03, the same pinned vLLM 0.8.5.post1 / V1 engine / bfloat16 / Flash-Attention backend, prefix caching off, sequential requests — and re-measured the floor there before any counterfactual: 20/20 null replays exact across ten configurations and both models (a gate, not a full 270-null floor; the C_H , C_M , C_{HM} arms it is compared against were measured on the 4090).

Protocol cost. Exactness is bought with throughput: the premises force strictly sequential serving, one request at a time, on one GPU. The null-replay inventory is the canonical 9,296 of Table 11 (dedicated plus attached, reconciled per artifact); the counterfactual inventory adds 1,380 replays at 460 interaction points (three per point), 13 census points and 180 stochastic replays in the round-4 cells, 294 single-flip C_H replays, 304 re-sampling replays with 401 a' sampling calls (pre-registration 21), and 1,617 census a' sampling calls (up to 8 seeds each); the V2 census chain (§7) adds 120 episodes, 117 nulls, 693 screening replays and the census arms in roughly 75 GPU-minutes; the training comparison adds 4,804 episodes (Act 2) plus the dose-matched replays; the fourth-cell registrations add 186 (pre-reg 40,

incl. 16 nulls) and 260 (pre-reg 42, incl. 20 nulls) rollouts, the latter in about 15 GPU-minutes on the H100. The chained experiment logs account for roughly 25 wall-clock hours on the single RTX 4090 (an under-count: the earliest runs predate chained logging). Dose-matching makes the replay tax concrete: at equal rollout budget, the interaction-corrected arm afforded 319 training episodes against the outcome-only arm’s 1,789 — a $5.6\times$ episode cost that any replay-based trainer must amortize.

Beyond determinism: the stochastic protocol. Real agent stacks violate every exactness premise (sampling temperature, flaky tests, batched serving). The protocol degrades gracefully rather than breaking: under stochastic replay each branch reward becomes a distribution, $C(d)$ a difference of means, and the measured null-replay distribution replaces the exact floor. The per-point recipe is: (i) estimate the null spread from n_0 null replays (the analogue of §4.1, now a variance rather than a maximum); (ii) size the per-branch replay count so that the minimal credit of interest exceeds the resulting standard error; (iii) cluster inference by task, as here. Exact screening-off becomes a test of $C_{HM} - C_M$ against the null band rather than a literal equality, and the coalescence lemma weakens to a coupling argument; what survives unchanged is the double-counting identity, which is algebraic and holds in expectation. The deterministic setting of this paper is the calibration case where all of this is measurable at zero variance — we deliberately paid throughput for that identifiability. We then executed this recipe (pre-registered, 180 replays): at temperature 0.8 with 12 seeds per arm, on the two highest- $|C_H|$ screened points and the three pressure breaks of the mt6 census. Declared outcome s1 held: at both screened points the stochastic $D = \bar{C}_{HM} - \bar{C}_M$ covers zero (at one of them exactly — every one of 36 replays returns the same reward); at 2/3 break points D excludes zero with the sign predicted by the deterministic taxonomy (api_router $D = +0.46$, CI $[0.38, 0.58]$; l_log_parser $D = -0.15$, CI $[-0.15, -0.13]$, against null spreads of 0.00–0.04), while the third (l_vending_machine, the weakest deterministic break at $|C_{HM} - C_M| = 0.09$) dissolves into the sampling noise. The deterministic census is thus not an artifact of greedy decoding: it predicts where the stochastic estimand separates from its null band.

G Training Acts 1–3 in Full

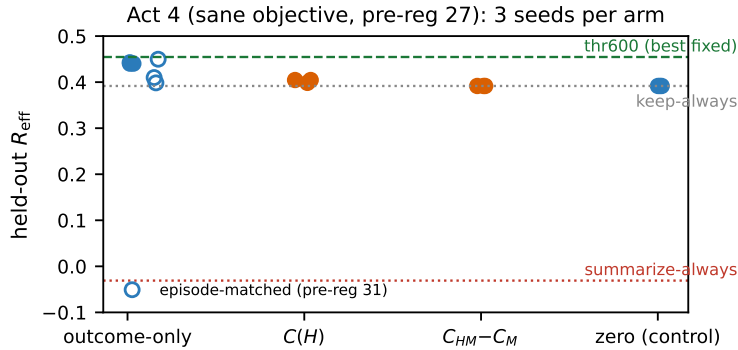


Figure 4: Act 4 (pre-regs 27, 31): held-out R_{eff} , 3 seeds per arm. Outcome-only (blue) beats the exact C_H arm (orange) in every seed; the interaction-corrected arm sits exactly on the untrained keep-always control. Hollow circles: the episode-matched outcome-only control (pre-reg 31), $\geq C_H$ at equal episodes in 3/3 seeds. Dashed: the expressible thr600 target; dotted: fixed attractors.

Act 1: seed-dependent collapse. With raw features and $\text{lr} = 0.5$, every trained arm collapsed into one of the two fixed policies, with the attractor determined by seed (outcome: k/k/s; C_H : s/s/k; $C_{HM} - C_M$: k/s/s). GATE 3 failed. Two hypotheses remained: the credit estimator is broken, or the optimization is. A pre-registered audit recomputed 60 logged credits by fresh two-greedy-replay: 58/60 sign agreement (mean discrepancy -0.0015).

Defining a harmful flip as logged-positive/recomputed-negative — a definition we state here, the registration specified sign agreement only — there are zero: the two disagreements are one benign flip (inventory_restock idx 0, $-0.084 \rightarrow +0.10$; logged as harmful, actually helpful) and one zero-to-negligible boundary case (c_temp_label idx 6, $0.0 \rightarrow -0.001$). The estimator is exonerated.

Act 2: stable optimization, no discovery. With the pre-registered fix (feature centering, $\text{lr} = 0.1$, gradient clipping), training is sane: gradients bounded, learned weights point the right way (negative bias, positive weight on context tokens, implicit decision thresholds at 2.5k–5.8k tokens). Yet no arm in any seed discovers the threshold policy: all non-collapsed runs converge to keep-always (held-out $R_{\text{eff}} = 0.237$ vs. thr600’s 0.398; Figure 4). Credit-driven arms also retain collapse risk the outcome-only arm avoids (C_H : 1/3 seeds; $C_{HM} - C_M$: 1/3; outcome-only and zero: 0/3) — a seed-count difference that is anecdotal at $n = 3$ (Fisher $p = 1.0$) and that we report without testing.

Diagnosis. Dose-matching in total rollouts means credit arms pay for replays with episodes: across 3 seeds the outcome arm saw 1,789 training episodes, C_H 786, $C_{HM} - C_M$ 319 (zero: 1,910). One might suspect the credit arms simply saw too little data. The state-visitation and reward-margin records rule this reading out and localize the failure in three steps. First, the learned policies are behaviorally inert: their implicit decision thresholds sit at 2,500–5,800 context tokens, but no decision in any arm or seed — 0 of 13,498, including the zero arm, de facto keep-always, the policy that maximizes context growth (1,910 episodes, cap 1,532 tokens) — and no held-out decision (cap 1,200 tokens) ever reaches that region: the learned thresholds can never fire, so every non-collapsed policy is keep-always in behavior. Second, in the states the training tasks do visit (thr600 fires above 600 tokens, well inside the visited range), the reward margin between thr600 and keep-always is statistically indistinguishable from zero: paired bootstrap on the training tasks gives $+0.024$, $\text{CI}_{95} [-0.029, +0.079]$, $P(\text{diff} \leq 0) = 0.19$ (re-derivable from the raw calibration artifact by the bundled `margem_calibacao.py`, which fails on drift) — whatever gradient signal exists on the training tasks is bounded by this margin, which is undetectable at this sample size; we note the interval admits margins (up to $+0.08$) that more episodes might resolve, so this second argument is an underpowered null and the diagnosis rests primarily on the state-visitation argument. Third, thr600’s held-out dominance comes from cumulative-cost structure: long held-out tasks accumulate prompt tokens under keep-always (e.g. l_vending_machine, 5,933 cumulative prompt tokens, $R_{\text{eff}} = -1.12$) — a cost regime the training distribution never makes consequential. The train/held-out split was alphabetical, fixed before training and before we knew this gap existed. The binding constraint is the training task distribution, not credit quality. This sharpens the design lesson: a cross-layer training benchmark must verify that its training tasks assign a nonzero reward margin to the states where candidate policies actually differ — otherwise credit fidelity is moot, and the failure will masquerade as an algorithmic one.

Act 3: verified margin, third attractor. We then ran the experiment this lesson demands (pre-registered with declared outcomes before any run). Stage A measured, for all 52 unique tasks in the project, the actual R_{eff} margin between thr600 and keep-always under two deterministic episodes per task, and selected the 16 tasks with positive margin (train/held-out split by alternating rank, fixed before training); held-out references: keep-always -1.181 , thr600 -0.746 , a verified margin of $+0.435$. Stage B re-ran all four arms $\times$ 3 seeds, dose-matched at 1,600 LLM calls, with the Act-2 optimization. The result is sharper than either prior act: every trained arm in every seed — outcome-only and both credit arms, 9/9 runs — converges to the same third attractor, summarize-always, scoring held-out $R_{\text{eff}} = -0.143$ with $R = 0.000$, while the zero arm (de facto keep-always) solves 45% of held-out tasks at $R_{\text{eff}} = -1.181$. The trained arms are right: on this pool of long, hard tasks under $\lambda = 25$, universal cheap failure is the true optimum of the declared objective — solving a task costs roughly its reward in tokens. Training worked; the objective was mis-specified. The declared outcome bin is s2 (all arms escape keep-always, no credit-vs-outcome ordering), and we own an omission: summarize-always is a fixed policy whose R_{eff} was computable from the stage-A machinery before stage B ran — the dominance of the third attractor was checkable analytically and we did not check it. The design lesson finalizes into its strongest

form: verifying a margin between two named policies is not enough — the margin must be verified against every fixed attractor expressible by the policy class (here summarize-always dominates both named references), and the cost weight λ must leave solving profitable, or every signal — outcome or credit, at any fidelity — will honestly optimize its way to failure. A secondary observation at matched call dose: the credit arms reached the same endpoint from roughly 9× fewer episodes (outcome 756–825 episodes per seed; $C_{HM} - C_M$ 82–87), consistent with credit buying per-episode signal at the price of replays, but on this landscape there was only one place to go.

H Additional Figures and Tables

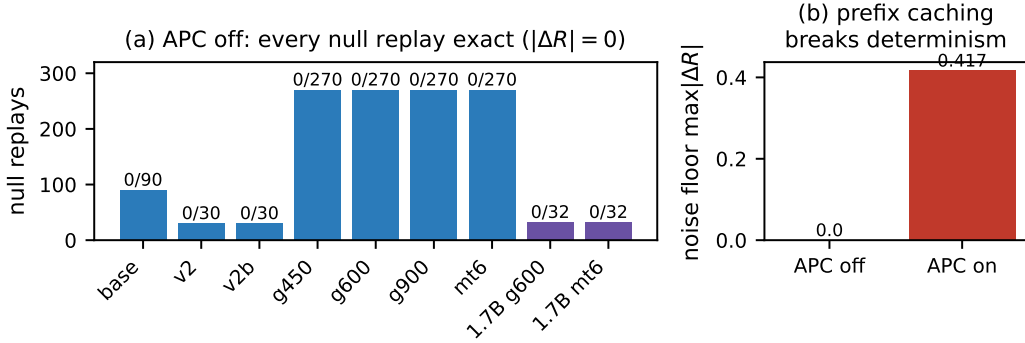


Figure 5: F1 — Measured noise floor. (a) Dedicated null-replay runs (Qwen3-4B blue, Qwen3-1.7B purple); together with nulls attached to interaction runs, the Qwen3-8B and MBPP+ replications, and the curated-pool runs, all 9,296 null replays are exact, floor = 0.0. (b) With vLLM automatic prefix caching on, the floor rises to 0.417 (7/270 inexact): greedy determinism breaks silently.

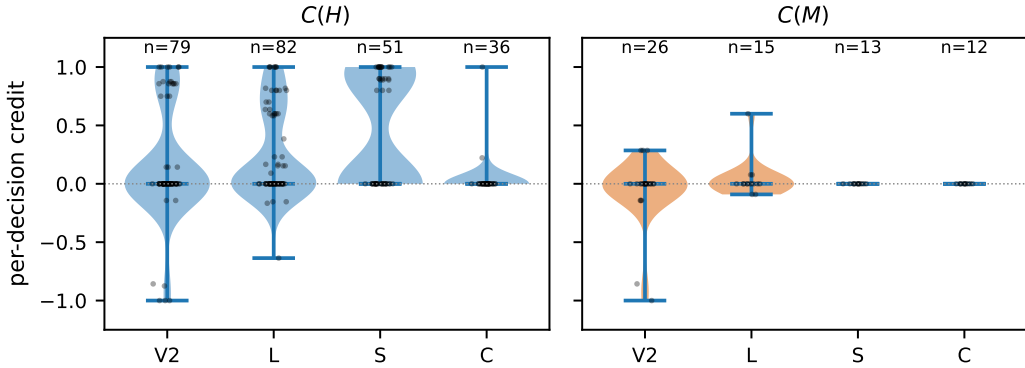


Figure 6: F2 — Per-decision credit by layer and stratum. C_H (248 points) and C_M (66 points) against an exact zero floor. Strata: V2/L information-destruction, S designed synergy, C controls.

I Critic Detail

Per-stratum detail for §5: on the combined L+V2 strata ($n = 161$, task-clustered Spearman) the GBM reaches 0.667 and the linear critic 0.675, against 0.708 for the position heuristic and 0.770 for $|\text{context}|$. Per stratum, L is a tie (GBM 0.747 vs. position 0.741), in V2 the heuristic wins (0.693 vs. 0.796), and S saturates for everything (AUROC 1.0). For detection

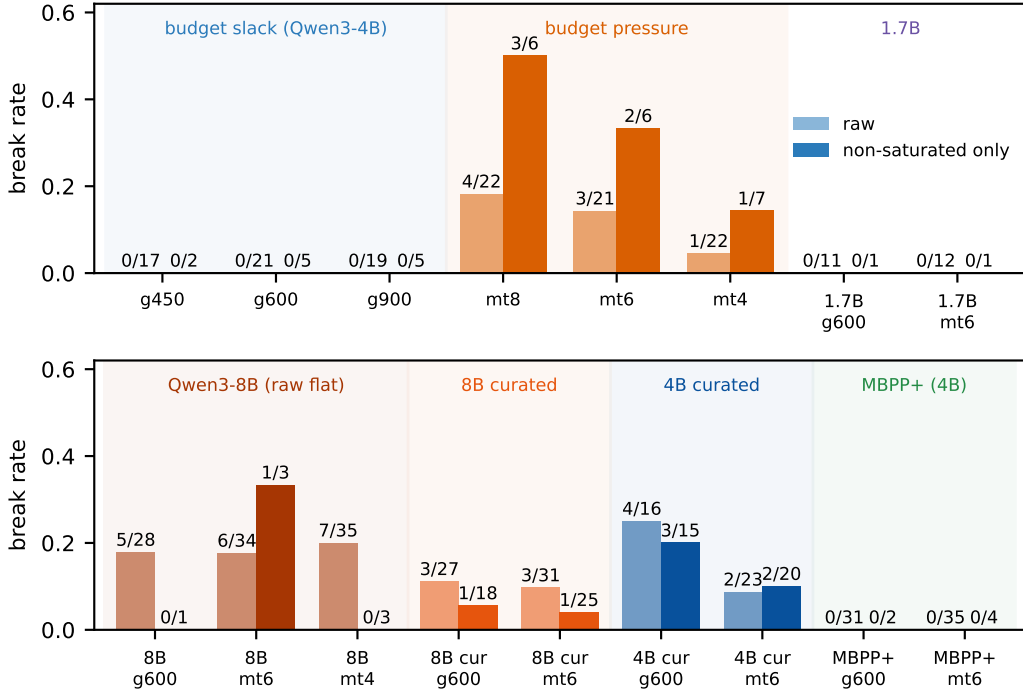


Figure 7: F3 — The slack/pressure contrast is itself a property of the triple: regime-dependent on the primary triple, flat or absent on others. Break rate of $C_{HM} = C_M$ per configuration, raw and conditioned on non-saturated points. Slack: 0/57 raw (0/12 non-saturated). Pressure: 8/65 raw (6/19 non-saturated, $p = 0.037$). Qwen3-1.7B (purple) replicates slack; its pressure null is explained by saturation (only 1 non-saturated point). Qwen3-8B (red) breaks raw screening at a flat rate across regimes (5/28, 6/34, 7/35), and the flatness survives the pre-registered curated pool (orange: 3/27 slack, 3/31 pressure); the symmetric control (dark blue) shows the 4B also breaks on the curated pool (4/16, 2/23) — the triple dependence of §8. MBPP+ (green, Qwen3-4B) replicates screening unconditionally (0/66).

($|C| > 0$) the GBM has a modest edge (AUROC 0.846 vs. 0.735 for position and 0.785 for the sign-inverted context-size heuristic) with overlapping CIs. The transfer test evaluated on the 94 MBPP+ C_M points (the only quantity with ground truth on both sides; C_H has none in the MBPP+ runs): Spearman 0.03 linear / -0.08 GBM, AUROC 0.30/0.50, MAE worse than a constant predictor.

J Pre-Registration Ledger

All confirmatory analyses were pre-registered in the project’s execution plan before the corresponding data were collected, with one dated exception. The ledger is an internal, same-team pre-specification, timestamped by git commits; for every item but one the registration commit precedes the corresponding run artifacts. The exception is item 10: the first slack census (g600) completed 34 minutes before the registration commit, so we treat g600 as discovery data and report the discovery/confirmation split in §4.3 (confirmation alone: 39/42). It is not a third-party registration, and minutes-to-hours gaps in a single-team pipeline cannot exclude unreported exploration before each commit (the pilot runs that preceded the formal ledger are preserved in the artifact tree). Its evidential force rests on the recorded failures, not on the registration mechanism itself. Table 9 lists all 43 pre-registrations with their outcomes, including the failed and the partial ones. Failed predictions are reported in the main text where they occur.

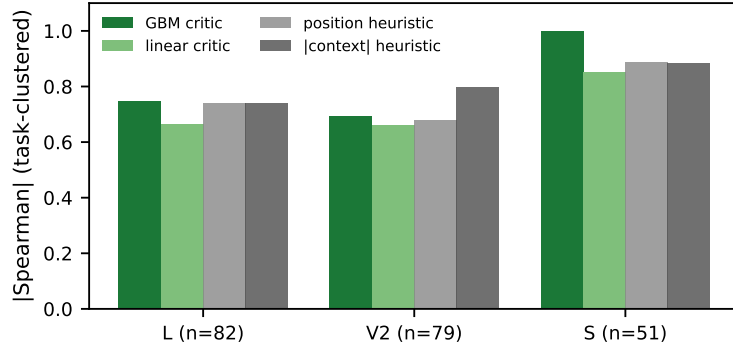


Figure 8: F4 — Critics vs. trivial heuristics per stratum (task-clustered $|\text{Spearman}|$). The critic never beats the best heuristic; C_H is structured by position and context size.

K Decision-Rule Audit

Table 10 audits the two-gate decision rule (census gate, Corollary 3; power gate, pre-regs 32/34/39) across every cell it touched. Gate entries are prospective where the rule predated the outcome (pre-reg 39) and retrospective elsewhere; every post-hoc entry was declared in the experimental diary before computation, after — and prompted by — adversarial review. Each gate has opened at least once and vetoed at least once; the conjunction (both open on the same cell) has not been observed. Whether it is attainable is a registrable question, not an assumption: the V2-4B row is a partial counter-example to structural anti-correlation (census open, power closed), and the $R_{\text{eff}}(\lambda^*)$ recomputation on the 8B cell rules out the estimand-mismatch mechanism there (Appendix N). Extending that recomputation to the stochastic cells shows the exact cross-check is specific to the near-deterministic cell, and distributionally so: every token-only crossing of the 0.10 scale lies within the same cell’s null-replay support in R_{eff} (39/39 on the 4B, 44/44 on Mistral, all ≤ 0.166 ; median permutation p against the null 0.18 and 0.41; $p < 0.05$ in 4/39 and 1/44, as exchangeability predicts). The stronger diagnostic: the null floor itself crosses the 0.10 scale in 30% (4B) and 60% (Mistral) of null replays — the registered scale sits below the R_{eff} noise floor of these cells, so a census run natively in R_{eff} on a stochastic cell needs its own null floor, mirroring §4.1.

L Null-Replay Reconciliation

Table 11 reconciles the canonical null-replay total (9,296, all exact) against every run artifact; a bundled script re-derives the table and fails if any count drifts. Excluded runs are listed with their reasons: the ASLR pre-fix runs (premises held, fidelity broken by an unsanitized memory address; root-caused and re-run) and the APC-contaminated run (premises violated by design, kept as the Figure 5b demonstration).

The ASLR incident in full. In a discarded pre-fix run of the first hardened pool (excluded from the 9,296: its pool failed the pre-registered curation rule, and its configurations were re-run after the fix), 2 of 432 nulls were inexact — the first fidelity breaks observed in this project (2/9,728 including every null ever run under the stated premises). Root cause, identified from the diverging trace: a memory address inside a Python repr in the pytest output (ASLR) entered the model’s live context, and greedy decoding diverged downstream. The sanitizer now normalizes addresses at the source (regression-tested), and the affected configurations were re-run from scratch. Two honesty notes: the 5,023 nulls collected before the fix were exact because their pytest outputs happened to contain no address-bearing reprs — measured, not guaranteed by sanitization; and “four curated-pool configurations” means 2 budget regimes $\times$ 2 pool halves. The lesson generalizes: replay-grade determinism requires auditing every byte that reaches the context, not just the serving stack.

The concurrency incident. A scheduling error in our own pipeline briefly ran two experiment chains against the same server, and the very first affected cell produced a reproducible non-exact null ($\Delta R = +0.67$ twice, diverging at the second retry call). Concurrent batching, like APC, is sufficient to destroy the floor; the contaminated runs were quarantined and re-run sequentially, and the incident is preserved in the run ledger.

The LLM-summarizer cell. A pre-registered cell replaces the rule-based summarizer with a greedy LLM summarizer (the served model summarizes its own history under a fixed prompt, temperature 0), re-running the full pipeline on the same 30 tasks: all 317 null replays (270 dedicated + 47 attached to the census) reproduced R exactly. The premise “greedy summarization is deterministic” was declared falsifiable in the pre-registration; it held. The cell’s census had low pivotal yield — 3 pivotal points out of 13 measured, 2 screened, 1 break (saturated) — because the semantic summary preserves the very constants whose destruction generates pivotal harness credit in this pool; we report the cell as underpowered for the screening question (its declared outcome bin) and informative for the floor question.

M Break-Point Anatomy (Descriptive)

This appendix reports the full descriptive trace anatomy of the curated-pool screening breaks (0 new rollouts; all from recorded replay traces). It is descriptive throughout: no count below is a pre-registered test.

The 8B’s six breaks. One uniform base and two patterns on top of it. Uniformly: every break is a keep→summarize flip at a write_file turn, the summary destroys essentially all critical constants in context (8/8, 5/6, 7/7, 10/10), and the harness flip alone is destructive. On top: either the sampled a' carries some constants and partially rescues the destroyed context (the joint flip lands between the single flips: h_hotel_folio, h_cargo_manifest — the former tracks the model, $C_{HM} = 0.08$ vs. $C_M = 0.00$; the latter the harness, $C_{HM} = 0.46$ vs. $C_H = 0.62$), or a' is itself beneficial and the destruction annuls exactly that benefit (the joint flip returns to the original reward: x_hours_bank, h_sku_validator — the one positive I is of this kind). The two recurring points are numerically identical across regimes — same four arm rewards, same destroyed constants, same free-divergence offsets, from different baseline trajectories — so the break is a property of the decision point (task $\times$ turn), not of the budget regime.

The 4B’s curated-pool breaks (pre-reg 19). The same anatomy adds a structural asymmetry: two of the 4B’s four unique break points (three of the six break instances, counting the recurring h_turnstile_fsm point) are restore-direction flips (summarize→keep: the baseline had already summarized, and the flip restores destroyed constants), a direction absent from the 8B’s breaks, and they follow a third pattern — individually beneficial or neutral flips whose joint application destroys the benefit ($I > 0$). Only one 4B point falls into an 8B pattern.

Audit details. The recurring h_hotel_folio point is numerically identical across regimes in all four arm rewards despite distinct baseline trajectories, and the recurring h_turnstile_fsm magnitude difference ($I = +0.77$ slack vs. $+0.31$ pressure) traces entirely to baseline quality — C_{HM} and all free-divergence offsets are identical across regimes. The I magnitude at a recurring point is therefore baseline-relative even when the break itself is stable.

N Design Brief for a Cross-Layer Credit Benchmark

The training negatives in the ledger are constructive: each names a property a benchmark must certify before a cross-layer credit comparison on it is interpretable. A pool qualifies only if it offers, per task:

1. Deterministic fractional reward (pre-regs 1, 27): a test suite scoring strictly inside $(0, 1)$ on the buggy state, deterministic at generation — otherwise the null floor is unmeasurable and no C is interpretable.

2. Controllable position of destroyable information (pre-regs 6, 16/20/25): the information a harness decision can destroy must be placeable relative to the decision point, with per-stratum calibrated recoverability; public benchmarks provide neither, which is why our external replications could test the floor and screening but not the regime contrast.
3. Verified attractor margins (pre-regs 32–34): for the declared policy class and cost weight, enough tasks (our gate: ≥ 10) on which the best conditional policy beats every expressible fixed attractor by a margin resolvable at the planned budget (≥ 0.10 in R_{eff}). Pre-reg 34 shows this must be verified, not assumed: on our V2 stack no cost weight yields even 3 such tasks, and training collapses regardless of learning rate or budget.
4. Cost weights scaled to the stack (pre-reg 32): λ grids do not transport across token-magnitude regimes; the benchmark must publish per-stack calibration data (fixed-attractor sweeps), not a single constant.
5. A screening census with registered accounting (pre-regs 29, 30): pivotal-point discovery, per-type screening rates, and the gate of Corollary 3 — with denominator and treatment of degenerate last-mover points fixed in advance, since the verdict can be sensitive to both (Table 6).

Items 1–2 make credit measurable; items 3–4 make training on it adjudicable; item 5 decides whether cross-layer correction is worth buying at all. Our designed pools satisfy 1–2 and 5 but — as pre-reg 34 established post hoc — not 3; to our knowledge no existing benchmark satisfies all five.

Item 3 is harder than it looks: after pre-reg 34 we attempted to construct a passing pool adversarially, with full knowledge of the mechanism (24 tasks: information poisoned against summarization paired with context-heavy tasks demanding it). Four design iterations, each calibrated against the real model, all failed the gate — and each failed through a distinct behavioral pathology rather than a task parameter: the model reads too little for the poison to bite the default policy; multi-stage repairs exceed its competence band entirely (all policies sink together); two-source synthesis collapses the pool; and in the final iteration, summarization mid-repair induced a degenerate loop (the model rewrites the same file without ever re-running tests, overflowing the context under every context policy). The margin of item 3 must be verified against model behavior, not estimated from task structure — exploratory, single-seed; the record is in the released ledger artifacts.

A follow-up on the final pool with a $2\times$ larger model (Qwen3-8B) sharpens the requirement rather than relaxing it. The looping pathology disappears, four tasks are solved by exactly the designed mechanism (keep overflows, summarize repairs, margin $+0.95$ each), and the conditional target strictly dominates every fixed policy in mean efficiency ($+0.16$) — the first landscape where it does. Yet the registered gate still vetoes (0/24): the conditional oracle commits at its first consulted decision and thereafter reproduces one fixed policy token-exactly per task, so its per-task margin over the best fixed policy is identically zero. A target that exercises only across-task conditionality can never pass a per-task-versus-best-fixed criterion; item 3’s margin must therefore be specified per attractor (against each fixed policy the optimizer can collapse into) and at the level the training signal observes — mean efficient reward — with any such revision registered before training, not after. Pre-reg 39 registered and executed that revision: the per-attractor gate opens (min margin 0.166 at $\lambda^* = 0.1$), and the mediation census then vetoed on new grounds — the 8B is near-deterministic at the paired decision points (462/463 resamples token-identical), so no a' exists and $C_{HM} - C_M$ has no contrast to measure. This adds a sixth requirement: model entropy at paired points must be verified before a cell is costed (report a' availability as a precondition diagnostic), and the census must screen in the estimand the training bills (R_{eff} at λ^*), not raw R — a margin priced in tokens is invisible to an outcome-level census. On the 39 cell this mechanism was then checked descriptively and did not bite: the census recomputed in $R_{\text{eff}}(\lambda^*)$ returns the identical pivotal set (token-only effects ≤ 0.033 , $5\times$ below the margin); the same recomputation on the stochastic 4B and Mistral cells is bounded by their own null-replay floor in R_{eff} (≤ 0.166), so the exactness is specific to the near-deterministic cell (Appendix K). What remains open — and registrable as this brief’s

first prediction — is whether the two gates anti-correlate structurally across cells: trainable margins may co-occur with exactly the low model entropy at paired points that empties the census. The V2-4B cell (census open, power closed) is a partial counter-example; the V1 post-hoc evaluation preserves the pattern (power opens on the trained pool, census closed retrospectively); a cell with both gates open has not been observed (Table 10).

O LLM Usage Disclosure

Large language models were used in this project in two roles. As the object of study: all measured agents (Qwen3-1.7B/4B/8B, Mistral-7B) are LLMs; their role is fully specified in the paper. As research assistants: LLM-based coding agents were used substantially to implement the measurement infrastructure and experiment scripts, to run and monitor pre-registered experiments, to draft and revise the manuscript text, and to run adversarial internal reviews of designs and drafts. All experimental designs, pre-registrations, scientific claims, and interpretations were decided and verified by the authors, who take full responsibility for the content; all reported numbers derive from the serialized artifacts checked by make reproduce, not from LLM output.

Claim	Evidence	Status
Replay noise floor is exactly zero	9,296/9,296 nulls exact, 3 models $\times$ 3 envs (Table 11)	confirmed (pre-reg 1)
Floor survives an LLM-in-the-loop summarizer	317/317 nulls exact (greedy summarizer cell)	confirmed (pre-reg 22)
Deterministic taxonomy predicts stochastic estimand	temp 0.8: 2/2 screened cover 0; 2/3 breaks exclude 0, predicted sign	confirmed (pre-reg 23, outcome s1)
Training negative survives a verified margin	9/9 trained runs find the true optimum of the mis-specified objective (cheap failure)	confirmed (pre-reg 24, outcome s2)
Under a sane objective, replay credit loses to outcome at matched dose	Act 4: arms separate 3/3 seeds; $C_{HM} - C_M$ (direct-effect signal) inert where $C_{Ha} = 0$; episode-matched control: outcome $\geq C_H$ 3/3 seeds at equal episodes; mechanism: credit density 0.39, grad-norm $5\times$ smaller	identified (pre-reg 27, 31, 41e)
Zero floor survives an agent-class change	V2 mini-SWE (multi-file, 5 decision types, plain-text protocol): 117/117 nulls exact	confirmed (pre-reg 29)
All five harness decision types are reward-pivotal	V2 screening: 114 pivotal points across all 5 types (incl. retry 2/2, test-schedule ΔR to -0.64)	confirmed (pre-reg 29)
The census gate has both branches	V1 stack: closed, confirmed by 4 training acts; V2 stack: outcome s3 (no type screens ≥ 0.90); gate opens under 3/4 accountings (0.50 measured w/ duals to 0.13 pivotal w/o duals, §7)	measured; accounting-sensitive (pre-reg 29)
V2 licensed training collapses onto a fixed attractor	8/9 learning runs converge token-exactly to summarize-always (held-out 0.831 vs. 0.889 dominant target), one outcome seed to keep-always (-0.092), before credit ordering is testable; first run aborted at calibration (λ does not transport across stacks)	identified (pre-reg 32–33, outcome s3)
Action envelope is a harness decision with estimand-level effects	JSON envelope: 0 writes in 437 calls; plain-text: sampler near-deterministic at temp 0.8 (87/114 fail), escalation registered	demonstrated (pre-reg 28a, 29b)
vLLM APC breaks greedy determinism	floor 0.417, 7/270 inexact	demonstrated
Slack screening-off, 4B original pool	0/57 raw; pivotal 53/56 = 0.946, CI95 [0.87, 1.00] task-clustered	internally consistent (pre-reg 10; endpoint re-specified post-hoc, Table 4)
Screening survives a' re-sampling	99/101 informative slack draws (CI95 [0.94, 1.00]); mt6 breaks persist 9/9	confirmed (pre-reg 21)
Screening completeness depends on the estimand	a'_s (summarized state): 24/31 slack, CI95 [0.69, 0.88]; de-screened points match the re-injection signature	measured; claim scoped (pre-reg 26)
V2 census verdict survives its estimand controls	re-draws: verdicts unanimous 36/46 under ≤ 8 draws/point; s3 + open gate hold in 7/7 schedules and under majority vote; under a'_s screening drops to 8/21 and all non-screened points turn exactly additive ($I = 0$, $C_H \neq 0$)	held; claim scoped (pre-reg 30 + 36)
Screening survives discovery/confirmation split	discovery (g600) 14/14; confirmation 39/42; strictly post-registration 27/30	confirmed (split forced by ledger audit)
Regime contrast not explained by selection composition	retained 8B points shallower than 4B's (turn 0.65 vs. 1.55); direction disfavors the contrast	supported (§8)
$C_H = -I$ at screened points	algebra (Prop. 1); exact in all measured slack points	identity
Pressure breaks recur at fixed points (4B)	8/65; same points across configs	point-level; task-clustered weak ($p = 0.125$)
Break rate monotonic in pressure	inverted U, Cochran–Armitage $p = 0.38$	failed (pre-reg 13)
Saturation mediates the window	conditioned rates only	descriptive (post-treatment) refuted
Screening-off is universal across models	1.7B: 0 raw breaks; 8B: breaks flat in both regimes	supported; positive characterization inductive
Shield is a property of the (model, harness, task) triple	curation + symmetric control; break sets point-level disjoint across models	failed (pre-reg 18)
Re-injection mechanism generalizes	23/24 annotated (original pool) vs. 0/15 exposed (curated), Fisher $p = 1.0/0.94$	
Screening replicates in a second environment	MBPP+: 11/11 pivotal (4B), 24/24 pivotal (1.7B); rest screens by construction; pressure cell empty	confirmed; pressure inconclusive (pre-reg 16, 20)
Screening replicates in a third environment	HumanEval+: 42/42 pivotal screened, 0 breaks; floor 1,294/1,294 exact	confirmed; pressure inconclusive (pre-reg 25)
Interaction-corrected credit trains a better harness	no arm learns thr600; margin $+0.024$, CI95 $[-0.029, +0.079]$	failed (pre-reg 9, 12); cause identified
Learned critic beats cheap heuristics	context-size heuristic wins overall	refuted; $C_H \approx$ position + size
Amortized credit transfers across environments	Spearman 0.03/ -0.08 , chance	refuted (pre-reg 16)
V2 collapse is a landscape property, not an optimizer artifact	analytic gat30 no λ gives ≥ 10 tasks with attractor margin ≥ 0.10 (max 3); probe at lr/5, budget $\times 2$ still collapses token-exactly, $\theta \approx 0$	held (pre-reg 34, both parts)
Structure and coupling not confined to	Mistral-7B under frozen V2 harness:	held, one cell (pre-reg 38)

Endpoint (one line)	Count	Note
Raw equality $C_{HM} = C_M$, slack, all measured points	57/57	headline mechanism; g450/g600/g900
Reward-pivotal sign test (pre-reg 10)	53/56	CI95 [0.87, 1.00] task-clustered; endpoint re-specified post hoc
Fourth cell $R(h', a) = R$, slack screened (pre-reg 40)	56/57	$I_{\text{fact}} = 0$; pivotal subset with $C_{Ha} = 0$: 36/37 (pre-reg 41, Table 2)
Discovery / confirmation split	14/14 / 39/42	g600 predates registration; strictly post-registration 27/30
a' re-sampling, 3 disjoint seed schedules (pre-reg 21)	99/101	stable outcome met; 79/81 on genuinely new a' ; mt6 breaks persist 9/9
Estimand control a'_s , summarized state (pre-reg 26)	24/31	partial: CI95 [0.69, 0.88]; all 7 de-screened show re-injection
Pressure regime, raw	8/65 breaks	shield cracks at recurring points

Table 4: Screening scoreboard: every screening-off count in the paper, with its endpoint. “Screened” counts points with $C_{HM} = C_M$; the pressure row counts breaks.

Model	Pool	Slack	Pressure	Reading
4B	designed (30)	0/57	8/65	breaks gated by pressure
1.7B	designed (30)	0/23	combined	slack replicates; pressure saturated
8B	designed (30)	5/28	6/34, 7/35	flat: regime-independent breaks
4B	curated (22)	4/16	2/23	breaks in both regimes
8B	curated (22)	3/27	3/31	flat again
4B	MBPP+ (60)	0/66	(11/11 pivotal)	screens unconditionally
1.7B	MBPP+ (60)	0/44	(24/24 pivotal)	screens unconditionally
4B	HumanEval+ (60)	0/76	(42/42 pivotal)	screens unconditionally
Mistral-7B	V2 multi-file (60)	22/43	non-screened	s3; outcome-level transport (pre-reg 38)

Table 5: Synthesis: raw screening-off break counts by (model, pool, regime). Pressure-gating is a 4B-on-designed-pool phenomenon; the 8B breaks at a flat rate inside its competence window; the external benchmarks screen everywhere. Counts are breaks/points; detail in Figure 7 (Appendix H).

	Non-screened mass	with duals	without duals
over measured points	24/48 = 0.50	(opens)	15/38 = 0.39 (opens)
over all pivotal points	24/114 = 0.21	(opens)	15/114 = 0.13 (closes)

Table 6: The V2 gate (threshold 0.20) under the four accountings left open: opens under three, closes under the strictest.

Act	Registered question	Identified failure	What it fixed for the next act
1	credit > outcome? (GATE 3)	optimization: lr 0.5 collapses every arm to a seed-dependent fixed policy; audit (58/60) exonerates the estimator	normalized features, lower lr, entropy floor
2	same, repaired optimizer	distribution: training tasks never visit states where policies differ (stable learning, no separation)	margin-verified task selection
3	same, verified margin	objective: mis-specified λ — all 9 runs correctly find its true optimum, universal cheap failure	analytic λ^* verified against all fixed attractors
4	same, sane objective	none — arms separate 3/3 seeds: outcome 0.440 > C_H 0.405 > corrected 0.392 = untrained	the closed branch of Prop. 2, confirmed

Table 7: The four training acts. Each failure was diagnosed by a pre-registered audit before the next act ran.

Method	counterfactual via replay	measures C_H	measures C_M	I on same trajectory	trains with I	structure across models tested
CHILL-Harness	✓ ^{off}	✓	—	—	—	—
CAR	✓	—	✓	— ^s	—	—
C3 v2	✓	(per agent)	—	—	—	—
Replay audit (Zhang, 2026)	✓	—	✓	—	—	~ ^m
Co-Harness / HASE	—	—	—	—	— [*]	—
Ours	✓	✓	✓	✓	tested [†]	✓

Table 8: T1 — positioning. ^{off}Checkpointed paired replay used offline to train an amortized estimator; disabled at decision time. ^sCAR measures step-step interaction (Shapley) within one layer, not across the model/harness boundary. ^{*}Joint harness+weights training without per-decision causal measurement. ^mObserves that counterfactual measurability is model-dependent, within a single layer; no cross-layer structure. [†]Tested with dose-matched controls and pre-registration; the outcome is an identified negative (§6).

Table 9: Pre-registration ledger. Failures are results, not omissions: #9 and #12 constitute the two-stage training negative (§6); #13 led to the competence-window analysis (§4.3).

Table 10: Decision-rule audit: every cell the two-gate rule touched, the state of each gate, what was paid, and whether the rule’s verdict matches the outcome.

Run	Model / pool	Dedicated	Attached	Inexact
g450, g600, g900	4B / original, slack	3×270	64, 61, 61	0
mt8, mt6, mt4	4B / original, pressure	3×270	61, 61, 61	0
v2, v2b	4B / pilot pools	2×30	16, 16	0
q17_g600, q17_mt6	1.7B / original	2×270	32, 32	0
q8_g600, q8_mt6, q8_mt4	8B / original	3×270	78, 78, 78	0
v4_g600, v4_mt6	8B / hardened v4 (post-fix)	2×216	—	0
v5_g600, v5_mt6	8B / v5 pool (post-fix)	2×180	—	0
v5cur_g600, v5cur_mt6	8B / curated union	—	66, 66	0
q4cur_g600, q4cur_mt6	4B / curated (symmetric control)	2×198	53, 53	0
mbpp_g600, mbpp_mt6	4B / MBPP+	2×540	107, 107	0
mbpp17_g600, mbpp17_mt6	1.7B / MBPP+	2×540	78, 78	0
he_g600, he_mt6	4B / HumanEval+	2×540	107, 107	0
ls600	4B / original, LLM summarizer	270	47	0
Canonical total		7,728	1,568	0
Excluded from the canonical total:				
v4_g600, v4_mt6 (pre-fix)	8B / hardened v4, ASLR leak	2×216	—	2
g600 (APC)	4B / original, APC on	270	68	7 + 2

Table 11: Run-by-run null-replay reconciliation. Canonical total $7,728 + 1,568 = 9,296$, all exact; including the pre-fix runs, 2/9,298 nulls ever run under the stated premises were inexact (both in one pre-fix run, §4.1). “Attached” nulls are the queue-floor checks run inside each interaction experiment.